AwkBeats Prep Station

Project documentation

AwkBeats Prep Station, version 1.0.0.

Operator: Awkward J (legal entity UNCONFIRMED). Brand site: https://awkbeats.com. There is no hosted app URL.

Repository: https://github.com/Awkward8787/awkbeats-prep-station

Branch: cursor/awkbeats-prep-station-e2e2

Commit: f99d3c6a38ad256e123fa13b50e0f7660369bd61

Date: 2026-10-21 (UTC). Status: Final (as shipped).

Contents

ABP-PRD — Product Requirements
ABP-TRD — Technical Requirements
ABP-FLOW — App Flow
ABP-DESIGN — Design Brief
ABP-DATA — Backend and Data Schema
ABP-PLAN — Implementation Plan

Combined glossary

Beat: one audio file plus its library row. Camelot: DJ key code, for example 11A is F# minor. Working BPM: bpmOverride if set, otherwise bpmDetected times bpmScale (0.5, 1, or 2). Lock: bpmLocked or keyLocked, which analysis will not overwrite. ISRC: 12-character recording code. userData: Electron's per-user directory, overridable with AWKBEATS_USER_DATA. PrepApi: the preload bridge on window.prep.

Combined open questions

ID	Question	Owner
Q-001	Legal entity for copyright and Apple Developer ID	Awkward J
Q-002	Keep the proprietary all-rights-reserved license?	Awkward J
Q-003	ISRC registrant prefix, or is the 12-character check enough?	Awkward J
Q-004	Approve the drawn AJ mark	Awkward J
Q-005	Will the app be sold, at what price, through which store? No price was provided.	Awkward J
Q-006	Support email	Awkward J
Q-007	Windows certificate: OV or EV	Awkward J
Q-008	Minimum macOS string for Electron 33.4.11	Awkward J
Q-009	Should remove ever delete the audio file? v1 does not.	Awkward J
Q-010	Wordmark lockup beyond the top bar	Awkward J
Q-011	Should catalog notes also be written into the audio comment?	Awkward J
Q-012	Relink a beat when its file moves?	Awkward J

Product requirements — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	Product Requirements Document
Document ID	ABP-PRD
Version	1.1
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. This is a desktop app. Brand site: https://awkbeats.com
Operator	Awkward J (legal entity UNCONFIRMED)

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1. Summary

AwkBeats Prep Station is a local desktop application for the producer Awkward J, and for other producers who prepare beats the same way. It keeps a searchable library of WAV, MP3, AIFF, and FLAC files, measures each beat's tempo to 0.01 BPM and its musical key in both standard notation and Camelot code, and writes title, producer, genre, mood,

tempo, key, tags, comments, copyright, optional ISRC, date, and cover art into the file's real tags. Version 1.0.0 ships as an unsigned macOS `.dmg` and a Windows `.exe`. Analysis and the catalog stay on the machine. Nothing in the app phones home.

2. Background and problem

Producers finish beats in a DAW and then lose track of key, tempo, status, and cover art across folders. Streaming stores and sample platforms want consistent tags (key, BPM, artwork at a large square size, ISRC when the beat is registered). Doing that by hand in a separate tag editor, a spreadsheet, and an image editor is slow and the numbers drift.

The owner's request, paraphrased from the build brief: a desktop app branded AwkBeats (awkbeats.com, producer Awkward J) to prepare and keep track of beats; import WAV, MP3, AIFF, and FLAC; analyze key and exact BPM locally; write real tags; attach square cover art with a 3000×3000 export; track status from idea through sold; export CSV; batch edit and batch analyze; ship macOS and Windows installers; avoid AGPL/GPL analysis libraries because the app may be sold.

Current alternatives the brief rejected: essentia.js and other GPL or AGPL detectors, and any workflow that uploads audio to a server. This version does not replace a DAW and does not sell beats itself.

3. Goals and non-goals

Goals

1. One local library the producer can search, sort, and filter without opening a spreadsheet.
2. Tempo and key measured on the machine, accurate on steady reference material to 0.01 BPM, with half-time and double-time and a hand override that can be locked.
3. Tags written into the formats distributors actually read: ID3v2 for MP3, RIFF INFO plus ID3 for WAV, ID3 for AIFF, Vorbis comments for FLAC, with embedded cover art.
4. A dark AwkBeats interface that is usable from the keyboard.
5. Installers for macOS and Windows produced by GitHub Actions and stored as workflow artifacts. Unsigned is acceptable for this version.

Non-goals

1. No user accounts, cloud library, or collaboration in v1.
2. No DAW editing, mixing, or stem export.
3. No beat store, checkout, or license PDF generator.
4. No Linux installer in v1. The app can run under Electron on Linux for development.
5. No image watermark burned into the cover. A text producer field is written as a custom tag. That is the shipped form of the optional watermark request.
6. No code signing or notarization until the owner supplies certificates (see ABP-PLAN).
7. No analytics, crash reporter, or auto-update server.

4. Success metrics

The app collects no analytics, by design (NFR-006). There is no traffic, conversion, or retention number to report, and none is invented here.

Proposed manual signals, not targets:

Signal	How it is checked	Status
Tempo on generated clicks, noise bursts, and soft pulses	`tests/bpm-key.test.ts` asserts 0.01 BPM	Shipped
All 24 keys on generated tones	Same file	Shipped
Tags round-trip on four formats	`tests/library-tags.test.ts`	Shipped
Both installers exist as CI artifacts	GitHub Actions workflow `Build installers`	Shipped as workflow; artifact proof is the Actions run for the shipping commit
A producer can finish import → analyze → write tags without a network	Manual run on the Electron window	Exercised in development on Linux with seeded beats

5. Audience and personas

Primary — Awkward J (working producer)

Role: beatmaker selling or leasing instrumentals. Technical comfort: high inside a DAW, impatient with tag editors. Context: studio desk, macOS or Windows, often offline. Needs: correct key and BPM, a cover that meets a 3000×3000 request, a status so a beat is not leased twice, a CSV for a catalog. Accessibility: full keyboard path, dark UI, no flashing motion. Age range: UNCONFIRMED.

Secondary — another producer using the same tool

Same needs. Default producer name and copyright template are settings, so the app is not hard-locked to one artist string, but the brand chrome says AwkBeats.

Not a persona in v1

Buyers of beats, playlist curators, and label admins do not get an account. They receive the tagged file and, if the producer sends it, the CSV.

6. User stories

US-001 Import

As a producer, I want to drop a folder of bounces on the window so that every WAV, MP3, AIFF, and FLAC becomes a library row.

- Given the library is open, when the producer drops files or a folder, or uses Import, then each new audio file is added once.
- Given a file path is already in the library, when it is imported again, then it is counted as a duplicate and not added twice.
- Given `Night Drive.jpg` sits beside `Night Drive.wav`, when the wav is imported, then that JPEG is stored as the beat's cover.

US-002 Find a beat

As a producer, I want to filter by status, genre, key, tag, BPM range, and date so that I can see what is ready in F# minor around 140 BPM.

- Given filters are set, when no row matches, the table says "No beats match these filters."
- Given a column header is clicked, the table sorts by that column.

US-003 Know the tempo and key

As a producer, I want each beat analyzed locally so that I see BPM to two decimals and a key like "F# minor" with Camelot "11A".

- Given a steady click track at 140.25 BPM, the detector returns 140.25.
- Given a tonal loop in F# minor, the detector returns `F# minor` and `11A`.
- Given the producer clicks Half-time, the working BPM becomes half of the detected BPM, not half of a previous half.
- Given BPM or key is locked, a later analysis does not replace it.

US-004 Write the file

As a producer, I want Write tags to file to store title, producer, genre, mood, BPM, key, tags, comments, copyright, ISRC, date, and the cover inside the audio file.

- Given a valid ISRC or a blank ISRC, the write proceeds. A malformed ISRC shows "ISRC must be 12 characters, like US-AWK-26-00001, or left blank."
- Given the file is read-only, the toast says "Tags could not be written. The file may be open in another app or read-only."

US-005 Cover art

As a producer, I want to crop a cover to a square and export at 3000×3000 so that a store's artwork rule is met.

- Given the source is smaller than 3000×3000 and 3000 is selected, the crop dialog says "This image is smaller than 3000×3000. Exporting at 3000 will upscale it."

US-006 Catalog

As a producer, I want a status, notes, batch edits, and a CSV so that I can run the catalog without a second tool.

- Status values are idea, in progress, ready, leased, and sold.
- CSV includes the working BPM and key, the detected values, locks, and file path.

7. Functional requirements

ID	Requirement	Priority	Status	Stories	Implementation
FR-001	Import WAV, MP3, AIFF/AIF, FLAC from files, folders, or drag and drop. Skip dotfiles. Reject files over 2 GiB.	Must	Shipped	US-001	`src/core/library/AudioFiles.ts`, `Importer.ts`, `src/renderer/App.tsx`
FR-002	Skip a path that is already in the library.	Must	Shipped	US-001	`Importer.importPaths`, `LibraryStore.findByPath`

ID	Requirement	Priority	Status	Stories	Implementation
FR-003	Search, sort, and filter by key, BPM range, genre, tag, date, and status.	Must	Shipped	US-002	`BeatQuery`, `Sidebar.tsx`, `TrackTable.tsx`
FR-004	Detect musical key as a name and a Camelot code, offline.	Must	Shipped	US-003	`KeyDetector.ts`, `Camelot.ts`
FR-005	Detect BPM to 0.01, offline, with half-time and double-time.	Must	Shipped	US-003	`BpmDetector.ts`, `BpmMath.ts`
FR-006	Hand override and lock for BPM and key. Locks survive re-analysis.	Must	Shipped	US-003	`LibraryStore.applyAnalysis`, `Inspector.tsx`
FR-007	Edit title, artist/producer, genre, mood, tags, comment, copyright, ISRC, date, status, notes.	Must	Shipped	US-004, US-006	`Inspector.tsx`, `BeatValidation`
FR-008	Write those fields into ID3v2 (MP3), RIFF INFO + ID3 (WAV), ID3 (AIFF), Vorbis comments (FLAC).	Must	Shipped	US-004	`TagWriter.ts`
FR-009	Import, crop to square, export at 1500 or 3000, embed artwork.	Must	Shipped	US-005	`SquareCrop.ts`, `dialogs.tsx` CropDialog, `TagWriter`
FR-010	Optional producer custom tag (`PRODUCER`). No burned-in image watermark.	Could	Shipped as text tag	US-004	`TagWriter.id3Tags`, Vorbis `PRODUCER`
FR-011	Status per beat: idea, in_progress, ready, leased, sold. Notes field.	Must	Shipped	US-006	`types.ts` `BEAT_STATUSES`
FR-012	CSV catalog export.	Must	Shipped	US-006	`CsvExport.ts`
FR-013	Batch edit and batch analyze.	Must	Shipped	US-006	`LibraryStore.batchUpdate`, `App.analyzeIds`
FR-014	Playback of the selected beat.	Should	Shipped	US-002	`beat://audio/<id>`, `Inspector.tsx`
FR-015	Keyboard shortcuts for search, import, export, analyze, write, remove, status, playback.	Should	Shipped	US-002	`App.tsx`, `dialogs.tsx`
FR-016	Settings: default producer, copyright template with `{year}`, default cover export size.	Should	Shipped	US-001	`DEFAULT_SETTINGS`
FR-017	Show the audio file in the OS folder.	Should	Shipped	US-004	`beat:reveal`
FR-018	Sidecar JPEG (`Beat.jpg` or `Beat.jpeg`) imported as cover.	Should	Shipped	US-001	`Importer.sidecarJpeg`

ID	Requirement	Priority	Status	Stories	Implementation
FR-019	Remove from library without deleting the audio file.	Must	Shipped	US-006	`LibraryStore.remove`
FR-020	Flag a row when the audio file is missing on disk.	Should	Shipped	US-002	`Beat.missing` computed on read

8. Non-functional requirements

ID	Requirement	Status	Where
NFR-001	Analysis and tagging libraries are MIT, BSD, Apache-2.0, or original code. No AGPL/GPL detector such as essentia.js.	Shipped	`scripts/check-licenses.mjs`, `src/core/analysis/`
NFR-002	Tests assert BPM and key against generated references. Tests are not weakened to pass.	Shipped	`tests/bpm-key.test.ts`
NFR-003	GitHub Actions builds a macOS dmg and a Windows exe and uploads them as artifacts. No GitHub Release.	Shipped	`.github/workflows/installers.yml`, `--publish never`
NFR-004	Dark AwkBeats UI. No placeholder copy and no developer notes in the interface.	Shipped	`src/renderer/styles.css`, `UserMessage.ts`
NFR-005	Renderer is sandboxed. Node is not exposed except through the preload bridge.	Shipped	`electron/main.ts` webPreferences
NFR-006	No network calls except the user opening https://awkbeats.com from the brand control, and the dev server in development.	Shipped	`SITE` constant, CSP in `src/renderer/index.html`
NFR-007	Keyboard access for the main tasks. Visible focus ring.	Shipped	`:focus-visible` in `styles.css`
NFR-008	Body text contrast meets WCAG AAA on the background. Muted labels meet AA.	Shipped	`--text-faint: #938d85` (see ABP-DESIGN)
NFR-009	Reduced motion disables transitions and animations.	Shipped	`@media (prefers-reduced-motion: reduce)`
NFR-010	Library writes are atomic (temp file, then rename).	Shipped	`LibraryStore.save`
NFR-011	English only in v1.	Shipped	UI copy
NFR-012	Window 1440×920, minimum 1100×700.	Shipped	`BrowserWindow` in `electron/main.ts`

Performance budget: the renderer JS bundle from the 1.0.0 Vite build is 188 KB (`dist/renderer/assets/index-*.js`) and the CSS is 20 KB, plus self-hosted IBM Plex subsets. Analysis is capped at the first 90 seconds (`TrackAnalyzer`). There is no page-load target because there is no web page.

Privacy: audio and the catalog stay in the user-data directory. No account, no cookie, no third-party script. COPPA, GDPR, and PCI are out of scope for v1 because the app does not collect customer data and does not take payment. If the owner later sells the installer, the storefront's privacy policy is a separate document (UNCONFIRMED, not in this repo).

Security: see ABP-TRD. The renderer cannot navigate to arbitrary URLs. `window.open` is denied.

Reliability: a failed file is reported in the import summary and does not abort the rest of the batch. Tag write failures are per file.

SEO: not applicable. There is no public HTML site in this repo.

Localization: English strings only. Dates are ISO `YYYY-MM-DD` sliced from the file mtime or an existing tag.

9. Content requirements

The interface is one window, not a website. There is no privacy page, terms page, or refund page inside the app. Legal copy for a future sale is an owner action (ABP-PLAN).

Surface	Purpose	Key copy
Window title	Identify the app	AwkBeats Prep Station
Brand control	Name and link	AWKBEATS / Prep Station. Opens https://awkbeats.com
Empty library	Tell the producer how to start	"No beats in the library yet" / "Drop WAV, MP3, AIFF, or FLAC files here, or import a folder."
No matches	Filter dead end	"No beats match these filters."
Missing file	File moved	"This file is missing from disk."
Analyze idle	Before analysis	"Analyze to measure tempo, key, and the waveform."
Tags written	Success	"Tags were written to the file."
Tags failed	Failure	"Tags could not be written. The file may be open in another app or read-only."
ISRC	Validation	"ISRC must be 12 characters, like US-AWK-26-00001, or left blank."
BPM	Validation	"BPM must be between 40 and 300."
Remove dialog	Confirm	"Remove from library" / "Remove the selected beats from the library? The audio files stay on disk."
Crop upscale	Warn	"This image is smaller than 3000×3000. Exporting at 3000 will upscale it."
Drop overlay	Drag feedback	"Drop audio to import"
Tag hint	Tags field	"Press Enter to add a tag."
Copyright help	Settings	"{year} is replaced when a beat is imported."

Tone: short, studio-floor, no jokes, no "lorem", no TODO, no "click here". The example ISRC `US-AWK-26-00001` is a format sample, not a registered code.

10. Monetization

The build brief says the app may be sold. Version 1.0.0 has no price, no payment provider, no tax handling, and no refund flow. Who would receive funds is UNCONFIRMED because the legal entity is UNCONFIRMED. The in-app copyright default is `© {year} Awkward J`, written onto beats, which is not a statement of the application's sale terms.

Status of payments: not applicable in this build.

11. Scope by release

v1.0.0 (this tree) — Shipped

Everything in the functional table marked Shipped: local library, analysis, tags, artwork, CSV, batch tools, unsigned installers, tests, docs.

Next — Planned

- Signed and notarized macOS build, signed Windows build, after the owner supplies certificates.
- Owner decision on price and storefront, if the app is sold.
- Optional burned-in producer mark on artwork, only if the owner asks. It was a nice-to-have and was not built.

Later — Planned, not committed

- Linux package.
- SQLite if the JSON catalog gets too large to rewrite on every edit.
- Waveform zoom and region analysis.
- Auto-update.

12. Assumptions and constraints

- The owner is the producer Awkward J, brand AwkBeats, site awkbeats.com. The installer icon is the iOS app icon (`build/icon.png`, gold A with a purple accent on black). The transparent file `build/logo.png` is the in-app mark. The earlier drawn AJ mark was removed.
- Budget, team size, and a launch date were not given.
- Hosting is the user's computer plus GitHub Actions for installers. There is no application server.
- Analysis must stay permissively licensed because the app may be sold.
- Installers in this version are unsigned.
- Node.js 22 is the CI runtime (`actions/setup-node` with `node-version: 22`).

13. Risks

Risk	Likelihood	Impact	Mitigation
Tempo octave errors on music without a clear pulse	Medium	Producer trusts a wrong BPM	Half-time and double-time, manual override, lock, confidence shown. Tests cover clicks, noise, soft pulses, and one drum loop.
Relative major/minor confusion	Medium	Wrong Camelot	Tonic salience term in `KeyDetector`. Tests cover all 24 keys on tonic-emphasized tones. Real mixes can still be ambiguous; the override exists for that.
Unsigned installers blocked by Gatekeeper or SmartScreen	High	Owner cannot hand the file to a customer easily	Documented signing steps. Artifacts are still produced.
Tag write damages audio	Low	Lost bounce	WAV rewrite keeps non-INFO chunks and tests that samples are unchanged. Other formats use ID3 or Vorbis block edits.
JSON catalog corruption	Low	Library list lost	Atomic rename. Audio files are not inside the JSON.

Risk	Likelihood	Impact	Mitigation
Commercial sale of a GPL detector	High if essentia were used	License conflict	Original detector plus MIT tag libraries. CI license check.

14. Stakeholders and approvals

Role	Who	Signed off
Owner / producer	Awkward J	UNCONFIRMED
Brand site	awkbeats.com	Referenced only. No claim that this app is published there.
Builder	Cursor Agent	This tree
Legal entity, Apple team, Windows certificate holder	UNCONFIRMED	Not signed off

15. Glossary

Term	Meaning
Beat	One audio file plus its library row.
Camelot	Key code used by DJs. `11A` is F# minor. `8B` is C major. Implemented in `Camelot.ts`.
Working BPM	`bpmOverride` if set, otherwise `bpmDetected * bpmScale`. `bpmScale` is 0.5, 1, or 2.
Lock	`bpmLocked` or `keyLocked`. Analysis will not replace a locked value.
ISRC	International Standard Recording Code, stored as 12 characters.
Prep Station	The product name of this app.
userData	Electron's per-user directory, overridable with `AWKBEATS_USER_DATA`.

16. Open questions

ID	Question	Owner	Blocked until answered
Q-001	What legal entity holds the copyright and would receive payment?	Awkward J	Signing identity, sale terms
Q-002	Is a proprietary "all rights reserved" license the one the owner wants?	Awkward J	LICENSE file
Q-003	Does the owner have an ISRC registrant prefix to validate against, or is the 12-character shape enough?	Awkward J	Stricter ISRC check
Q-004	Resolved. The owner supplied the iOS app icon and the transparent logo. They replace the drawn AJ mark.	Awkward J	None
Q-005	Will the app be sold, and at what price, through which store?	Awkward J	Monetization. No price was provided.
Q-006	Contact email for support?	Awkward J	Not shown in the app

17. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Initial PRD reconciled to the shipped 1.0.0 tree.	Owner build brief.

Date	Version	Change	Why
2026-10-21	1.1	Installer and in-app marks are the iOS logo. `build/icon.svg` is gone.	Owner supplied `icon.png` and the transparent logo.

18. Sources

- Owner brief in the task that created this repo (Awkward8787/awkbeats-prep-station).
- `package.json` version 1.0.0, `LICENSE`, `README.md`.
- `src/core/library/types.ts`, `UserMessage.ts`, `BeatRules.ts`, `Importer.ts`, `AudioFiles.ts`.
- `src/core/analysis/BpmDetector.ts`, `KeyDetector.ts`, `BpmMath.ts`, `Camelot.ts`, `TrackAnalyzer.ts`.
- `src/renderer/App.tsx`, `styles.css`, `dialogs.tsx`.
- `.github/workflows/installers.yml`.

Technical requirements — AwkBeats Prep Station

Document control

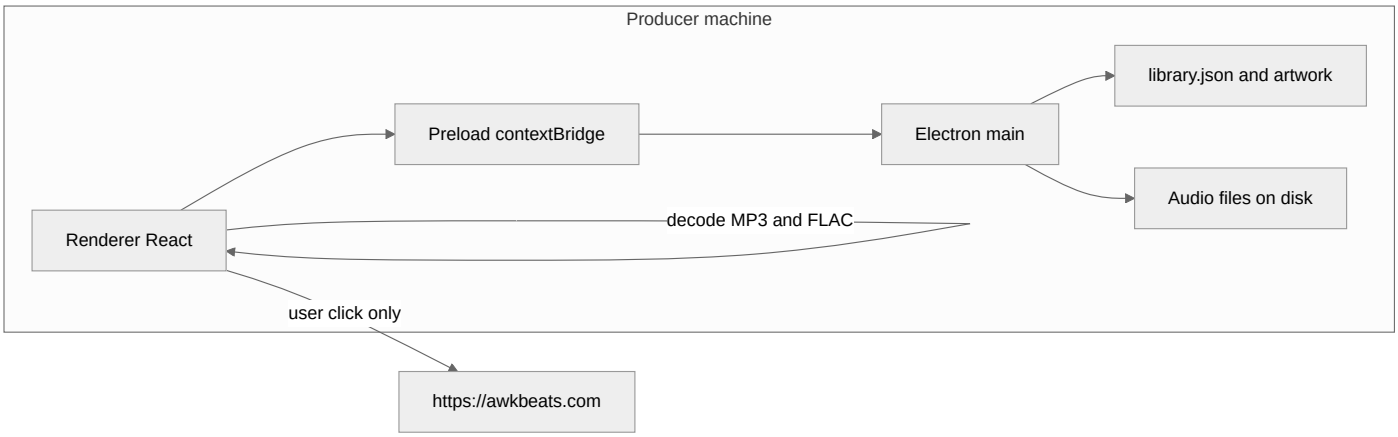
Field	Value
Project	AwkBeats Prep Station
Document title	Technical Requirements Document
Document ID	ABP-TRD
Version	1.1
Status	Final (as shipped)
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Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Desktop app. Brand site: https://awkbeats.com
Operator	Awkward J (legal entity UNCONFIRMED)

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1. Architecture

The app is a single Electron process plus one sandboxed renderer. There is no application server.



Process split. The renderer never reads the filesystem directly. The main process owns the library and the tag writer.

A request (an IPC call) flows like this. The renderer calls `window.prep`, which is the object exposed by `electron/preload.ts`. The preload uses `ipcRenderer.invoke` with a fixed channel name. `electron/main.ts` handles that channel, calls a class in `src/core`, and returns JSON-serializable data. Audio bytes for MP3 and FLAC analysis travel back to the renderer because Chromium already contains those decoders. WAV and AIFF are decoded in the main process with `PcmDecoder`.

The `beat://` scheme is registered as privileged before `app.ready`. `beat://audio/<id>` and `beat://cover/<id>` are handled with `net.fetch(pathToFileURL(...))` so the renderer can play audio and show art without a raw `file://` path in the page.

2. Tech stack

Versions are the ones installed for the 1.0.0 lockfile.

Layer	Technology	Version	Why	Alternatives rejected
Language	TypeScript	5.9.3	Shared types between main and renderer	Plain JS
Desktop shell	Electron	33.4.11	Mature installers, Chromium decoders for MP3/FLAC	Tauri (no Chromium decoder; would need another permissive decoder)
UI	React	18.3.1	Component model for the three-pane window	No router; one window
Styling	CSS file	`src/renderer/styles.css`	Small surface, no utility framework	Tailwind not needed
Fonts	IBM Plex Sans and Mono via fontsource	5.2.x	OFL, self-hosted	System fonts only would drift between macOS and Windows
Renderer build	Vite	6.4.5	Fast dev server	webpack
Main and preload build	esbuild	0.24.2	Bundle CJS, externalize electron and music-metadata	tsc emit
Test	Vitest	2.1.9	Same TS path aliases	Jest
Package	electron-builder	25.1.8	dmg and NSIS	Manual zip
Tags write	node-id3	0.2.9	MIT ID3v2 writer	Any GPL tagger
Tags read	music-metadata	7.14.0	MIT, CJS	—
Key and BPM	Original TypeScript	—	Permissive, testable	essentia.js (AGPL), aubio (GPL)

Layer	Technology	Version	Why	Alternatives rejected
Catalog	JSON file	schema version 1	No native module	SQLite / sql.js (wasm packaging deferred)
CI	GitHub Actions	checkout/setup-node/upload-artifact v4	macOS and Windows runners	Local cross-compile

Node in CI is 22. The development machine that produced this tree used Node 22.23.3 and npm 10.9.9.

State management is React `useState` in `src/renderer/App.tsx`. There is no Redux or router.

Linting: `tsc --noEmit` is the type gate. There is no ESLint config in v1.

3. Repository structure

```
electron/main.ts          window, menu, IPC, beat:// protocol
electron/preload.ts       PrepApi bridge
src/core/analysis/        Fft, BpmDetector, KeyDetector, Camelot, BpmMath, TrackAnalyzer
src/core/audio/           PcmDecoder, ReferenceAudio (tests and generators)
src/core/library/         LibraryStore, Importer, BeatRules, CsvExport, SquareCrop, types
src/core/tags/            TagReader, TagWriter, TagCommit, TagTypes
src/renderer/             React UI, styles, index.html CSP
tests/                    bpm-key and library-tags
tests/fixtures/silence.flac
scripts/dev.mjs           Vite + esbuild watch + Electron
scripts/check-licenses.mjs
scripts/seed-demo.ts      local demo library, not shipped in the installer
build/icon.png, logo.png, icon.ico, icon.icns, icons/
electron-builder.yml
.github/workflows/installers.yml
docs/                     this documentation set
```

`dist/` and `release/` are build outputs and are gitignored.

4. Environments

Environment	URL	How it runs	Difference
Local dev	`http://localhost:5173` inside Electron, set as `VITE_DEV_SERVER_URL`	`npm run dev`	Vite HMR. CSP allows that origin.
Local production build	`file://` to `dist/renderer/index.html`	`npm run build && npm start`	No dev server.
Installer	none until the owner distributes the artifact	`npm run dist:mac` or `dist:win`, or CI	asar archive. Unsigned.
Staging	Not applicable	No staging host	—
Hosted production	Not applicable	The app does not deploy to a server	—

`AWKBEATS_USER_DATA` points the library at a chosen folder. Unset, Electron uses its default userData path for app id `com.awkbeats.prepstation`.

5. Configuration and secrets

No secret is required to build or run the unsigned app. Signing secrets are owner-supplied later and must never be committed.

Name	Purpose	Required	Default	Where set	Where read	Rotation
`AWKBEATS_USER_DATA`	Override library directory	Optional	Electron userData	Process environment	`electron/main.ts` before ready	Change the path; move `library.json` and `artwork/` together
`VITE_DEV_SERVER_URL`	Dev renderer origin	Dev only	Unset in production	`scripts/dev.mjs`	`electron/main.ts` `createWindow`	Restart dev
`CSC_LINK`	Base64 Apple or Windows signing cert	Optional, signing only	Unset	GitHub Actions secret	electron-builder, not application code	Revoke the cert with the vendor, replace the secret
`CSC_KEY_PASSWORD`	Password for `CSC_LINK`	With `CSC_LINK`	Unset	GitHub Actions secret	electron-builder	Same as the cert
`WIN_CSC_LINK` / `WIN_CSC_KEY_PASSWORD`	Windows-specific override	Optional	Unset	GitHub Actions secret	electron-builder	Same
`APPLE_API_KEY`, `APPLE_API_KEY_ID`, `APPLE_API_ISSUER`, `APPLE_TEAM_ID`	Notarization	Optional	Unset	GitHub Actions secret	electron-builder notarize	Rotate the API key in App Store Connect
`CSC_IDENTITY_AUTO_DISCOVERY`	Stop the builder from grabbing a random local cert	Set to `false` in CI	false in the workflow	`.github/workflows/installers.yml`	electron-builder	Leave false until a signing job exists
`GITHUB_TOKEN`	Default Actions token	Present on Actions	—	GitHub	Not read by app code. `publish: null` and `--publish never` stop electron-builder from creating a Release	GitHub rotates it

There is no `.env` file. Do not add signing material to one.

6. Third-party integrations

No payment, email, auth, analytics, storage, CDN, or AI API is called.

Integration	Account	Endpoints	Auth	Failure
awkbeats.com	The owner's site. Account ID UNCONFIRMED.	`https://awkbeats.com` via `shell.openExternal` when the brand control is activated	None	OS browser handles failure. The library still works.
GitHub Actions	Repo Awkward8787/awkbeats-prep-station	Workflow artifacts only	GitHub permissions: `contents: read`	Installer job fails if `release/*.dmg` or `release/*.exe` is missing
music-metadata, node-id3	npm packages, not accounts	Local function calls	None	Tag read/write throws; UI shows the tag-failed message or the import failure count

Sandbox versus production: not applicable. There is one build.

7. Security

Authentication: none. The person at the keyboard is the user. The library is as private as the OS user account.

Authorization: IPC channels are a fixed list in `electron/preload.ts`. The renderer cannot pass a channel name of its own choosing through the bridge. Paths come from the file dialog or from `webUtils.getPathForFile` on a dropped file.

Input validation (`BeatValidation`):

- ISRC: empty, or `/^[A-Z]{2}[A-Z0-9]{3}\d{7}$/` after removing hyphens and spaces and uppercasing. Stored canonical, 12 characters.
- BPM override: 40 through 300 inclusive (`BpmMath.inRange`).
- Tags: trim, collapse whitespace, case-insensitive de-dupe, max 24 tags, 32 characters each.
- Status: one of `idea`, `in_progress`, `ready`, `leased`, `sold`.
- Comment max 4000 characters. Copyright max 400.
- Audio import: extension allow-list, size greater than 0 and at most 2 GiB, skip dotfiles.
- Cover export size: 1500 or 3000 only.

CSRF and CORS: not applicable to a local window. The `beat` scheme sets `corsEnabled` so the renderer can fetch it. Navigation: `setWindowOpenHandler` returns deny. `will-navigate` allows only the dev server origin and `file:`.

Rate limiting: not applicable. There is no public endpoint.

CSP in `src/renderer/index.html`:

```
default-src 'self' beat:;
script-src 'self';
style-src 'self' 'unsafe-inline';
img-src 'self' data: blob: beat:;
media-src 'self' blob: beat:;
connect-src 'self' beat: http://localhost:5173 ws://localhost:5173;
```

`style-src 'unsafe-inline'` is there because the cover well sets a background image in an element style and the crop stage positions an image with inline coordinates. There is no HSTS header because the document is not served over HTTPS. The brand site is opened in the system browser.

Secret storage: none in the app. Dependency audit: `npm run licenses` fails on GPL-family licenses in production packages. That check passed on the 1.0.0 tree. File permissions: the library directory inherits the user's umask. Known accepted risk: an unsigned binary will be flagged by Gatekeeper and SmartScreen. Another accepted risk: `connect-src` includes localhost so the dev build can load Vite. The production window does not start a server, but the CSP text is the same HTML file.

8. Privacy

Data	Where	Retention	Third party
Audio the producer imports	Original path on disk. The app does not copy the audio into userData.	Until the producer deletes the file	None
Catalog	`library.json` in userData	Until the producer removes beats or deletes the folder	None
Cover JPEGs	`userData/artwork/<id>.jpg`	Deleted when the beat is removed from the library	None

Data	Where	Retention	Third party
Cookies, localStorage, IndexedDB	Not used	—	—

`library.json` can contain file paths, which may include the OS username. That file stays on the machine. The app does not ship a logger that uploads.

9. Performance

Production renderer build measured on this tree:

Asset	Size
`dist/renderer/assets/index-*.js`	190.62 kB (Vite, gzip 61.32 kB)
`dist/renderer/assets/index-*.css`	16.59 kB (Vite, gzip 3.64 kB)
IBM Plex woff2 subsets	about 8–16 KB each, weights imported: sans 400/500/600, mono 500
`dist/main/index.js`	668 KB (node-id3 bundled; music-metadata external)
`dist/preload/index.js`	4 KB

Code splitting: one renderer chunk. The window is one screen. Caching headers: not applicable (file load). Images: covers are JPEG written by the crop dialog at quality 0.92, or the original sidecar bytes. Fonts are self-hosted woff2. No CDN.

Analysis reads at most the first 90 seconds. Onset detection uses hop 32. Key detection uses FFT size 16384 and hop 8192. The waveform stored on the beat is 160 peaks, each 0–1 with three decimal places.

10. Accessibility

- The search field has a label ("Search the library") that is visually clipped with the `.sr` class and tied to `library-search`. Other fields have visible labels.
- Focus: `:focus-visible` is a 2px solid `#f3d7a4` ring, 2px offset.
- Keyboard: listed in `ShortcutsDialog` and implemented in `App.tsx`. Status keys 1–5 map onto `BEAT_STATUSES` order. Shortcuts do not fire while an input, textarea, or select is focused, except Escape closes a dialog.
- Dialogs close with Escape, Cancel, or Close. Focus is not trapped in a WAI-ARIA modal loop in v1 (see debt).
- Contrast: see ABP-DESIGN. Body text on `#0c0d10` is 17.1:1. Muted `#938d85` on `#2a2418` is 4.50:1, which meets WCAG AA for normal text.
- Reduced motion: all transitions and animations are disabled.
- The play button is 36×36 CSS pixels. Table rows are 44px tall. Filter rows are about 32px, under a 44px target. Noted as debt.
- Color is not the only status signal. The status word is also printed.
- There is no skip link. The window has no repeated web header to skip.

11. SEO and sharing

Not applicable. The app has no public routes, sitemap, robots.txt, canonical URL, or Open Graph tags. The HTML title is "AwkBeats Prep Station". The installer icons are `build/icon.png` (1024) and `build/icon.ico`.

12. Build and deploy

Commands:

```
npm ci
npm test
npm run typecheck
npm run licenses
npm run build
```

`npm run build` runs esbuild for `electron/main.ts` and `electron/preload.ts` (CJS, electron external, music-metadata external on main) and `vite build` with root `src/renderer`, `base: './'`, outDir `dist/renderer`.

Installers:

```
npm run dist:mac
npm run dist:win
```

Both pass `--publish never`. CI (`.github/workflows/installers.yml`) runs test, typecheck, and licenses on `ubuntu-latest`, then `dist:mac` on `macos-latest` and `dist:win` on `windows-latest`. Artifacts:

- `AwkBeats-Prep-Station-macOS` from `release/*.dmg`
- `AwkBeats-Prep-Station-Windows` from `release/*.exe`

`if-no-files-found: error`. Permissions are `contents: read`. `publish: null` in `electron-builder.yml`.

There is no cache to purge and no server to roll back. Rollback of an installed copy is the OS uninstall plus installing the previous artifact. Rollback of the repo is `git revert`.

Post-build check: `release/` contains the dmg or exe; the app launches; the unit tests still match known tempos.

13. Testing

Framework: Vitest 2.1.9, `vitest.config.ts`, timeout 30 seconds, alias `@core` to `src/core`.

`tests/bpm-key.test.ts` asserts:

- FFT magnitude peak location.
- Camelot parse and format, including F# minor = 11A and flats mapping onto sharps.
- `BpmMath` half, double, range, and working tempo.
- Click tracks at 72.5, 90, 96, 120, 128.05, 140.25, 150, and 174 BPM, each within 0.01.
- Noise bursts at 100 BPM within 0.01.
- Soft pulses at 110 BPM within 0.01 and confidence greater than 0.8.
- An 80 BPM click is not reported as 160.
- All 12 major and 12 minor keys on `ReferenceAudio.keyedTone`.
- A drum loop at 96 BPM in F# minor returns 96.00 and `11A`.

`tests/library-tags.test.ts` asserts PCM round-trip error under 1/30000; MP3, WAV, AIFF, and FLAC tag round-trip including decimal BPM, Camelot, tags, ISRC, and artwork; WAV audio samples unchanged after a tag rewrite; import, duplicate skip, lock surviving re-analysis, filters, CSV, ISRC validation, crop math, and batch status.

Run: `npm test`. CI status: the workflow file is in the repo. The run URL is the Actions run for the shipping commit (see ABP-PLAN).

Manual QA: launch, import, read key and BPM, edit metadata, open crop at 3000, write tags, export CSV. Screenshots of the library, the analysis card, the metadata editor, and the crop dialog were taken on a Linux Electron window during development. macOS and Windows GUI launch is not part of the workflow. The runners prove the installer files exist.

Known gaps: no automated UI test. No test that Chromium's MP3 decoder matches a fixture BPM. The detector itself is tested on PCM. No installer GUI test on this Linux tree.

14. Monitoring and logging

Electron and Chromium write their default stderr. The app does not log file contents, ISRC lists, or keystrokes to a server. A packaged app's logs stay in the OS console. Alerts and uptime checks are not applicable. There is no service to ping.

15. Browser and device support

Target	Support
macOS arm64 and x64	dmg via electron-builder. Exact minimum macOS is whatever Electron 33.4.11 declares (UNCONFIRMED as a marketing string).
Windows x64	NSIS exe. Per-user install, directory changeable, not one-click.
Windows arm64, Linux packages	Not built
Web browsers	Not a target. The renderer is Chromium inside Electron 33.
Phone	Not a target. Below 980px CSS pixels the layout collapses to two columns. That is for a small desktop window.

16. Technical debt and known issues

Item	Severity	Proposed fix
Unsigned installers	High for distribution	Owner certificates, then a signing job.
Dialogs are not focus-trapped	Low	Add a focus trap when a modal opens.
Some hit targets under 44px	Low	Increase `.status-item` padding if an audit requires a 44px target.
`style-src 'unsafe-inline'`	Low	Move cover and crop positioning to CSS variables set from classes.
Catalog is one JSON rewrite per edit	Low until thousands of beats	Migrate to SQLite with the same `Beat` fields.
music-metadata parses BPM as an integer	Low, already handled	`TagReader.readNative` keeps the decimal. Do not remove that path.
Half-time of a mix without a clear kick can still be the tempo the producer wants	Medium, product	The UI offers half, double, and a typed BPM.
`concurrently` and `wait-on` are unused devDependencies	Low	Remove on a later cleanup.
React and fontsource are production dependencies even though Vite bundles them	Low	Move to devDependencies to slim the asar. music-metadata must stay a production dependency. node-id3 is bundled into the main file.
Search label is screen-reader only	Low	The field is the only text input in the top bar. Add a visible label if a later audit wants it.

17. Glossary

Term	Meaning
IPC	Electron `ipcMain.handle` / `ipcRenderer.invoke` channels listed in ABP-DATA.
asar	Electron archive of `dist/**` and `package.json`.
NSIS	The Windows installer technology electron-builder uses.
PrepApi	The TypeScript type of `window.prep` in `electron/preload.ts`.

18. Open questions

ID	Question	Owner	Blocked
Q-001	Legal entity for the Developer ID certificate	Awkward J	Signed macOS build
Q-007	Which Windows certificate vendor, OV or EV?	Awkward J	SmartScreen reputation. EV is the one that trusts immediately.
Q-008	Confirm Electron's declared minimum macOS for the 33.4.11 binary before publishing a support line	Awkward J	Support page, which does not exist yet

19. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Initial TRD for the 1.0.0 Electron tree.	Owner build brief. Muted text color raised to `#938d85` so small labels meet WCAG AA.
2026-10-21	1.1	Installer icons are `build/icon.icns`, `build/icon.ico`, and `build/icons/`.	Owner supplied the iOS icon.

20. Sources

- `package.json`, installed versions listed in section 2.
- `electron/main.ts`, `electron/preload.ts`, `electron-builder.yml`.
- `src/renderer/index.html`, `vite.config.ts`, `vitest.config.ts`, `tsconfig.json`.
- `.github/workflows/installers.yml`, `scripts/check-licenses.mjs`, `scripts/dev.mjs`.
- `tests/bpm-key.test.ts`, `tests/library-tags.test.ts`.
- Bundle sizes from `du` on `dist/` after `npm run build` on 2026-10-21.

App flow — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	App Flow
Document ID	ABP-FLOW
Version	1.1
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. One desktop window. Brand site: https://awkbeats.com
Operator	Awkward J (legal entity UNCONFIRMED)

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1. Screen inventory

There is no URL router. Screens are the main window plus dialogs. Auth is never required.

Screen	Path	Purpose	Entry	Exit	Component
Library	window	List, filter, inspect the selected beat	Launch	Quit	`src/renderer/App.tsx`
Settings	dialog	Producer name, copyright template, default export size	Settings button, Ctrl/⌘ comma, macOS app menu	Save settings or Cancel	`dialogs.tsx SettingsDialog`
Shortcuts	dialog	Shortcut list	Shortcuts button, Ctrl/⌘ /	Close or Escape	`ShortcutsDialog`

Screen	Path	Purpose	Entry	Exit	Component
Batch edit	dialog	Apply status, genre, mood, artist, tags, copyright to the selection	Batch edit in the inspector	Apply or Cancel	`BatchDialog`
Remove confirm	dialog	Confirm removal from the library	Remove, or Ctrl/⌘ Backspace	Remove or Cancel	`ConfirmDialog`
Crop cover	dialog	Square crop and export size	Adjust crop	Use cover or Cancel	`CropDialog`
awkbeats.com	external browser	Brand site	Brand button	OS browser	`shell.openExternal`

Below 980px CSS width, the rail hides while a beat is selected (class `app.inspecting`) and the inspector hides when nothing is selected. The default window is 1440×920, so all three columns show.

2. Navigation

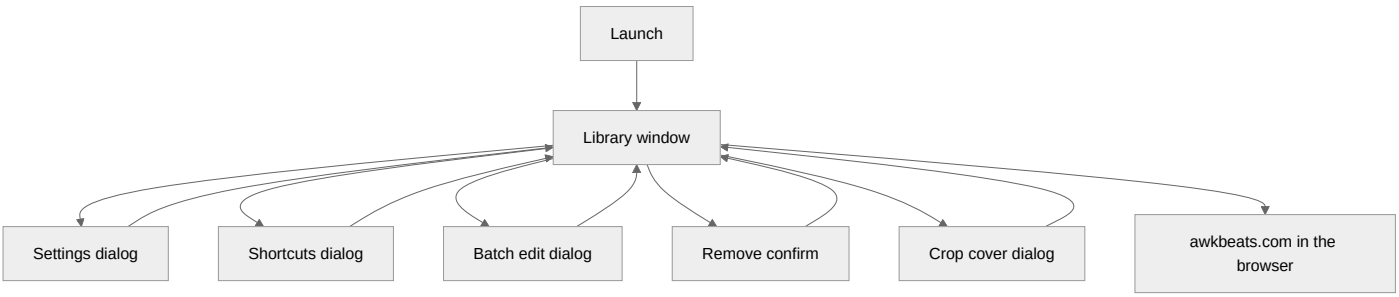
The top bar holds the brand button, the search field, Import, and the count of selected rows. The left rail holds status filters, genre, key, tag, BPM range, and dates. The center is the table. The right inspector is the selected beat. The footer holds the visible count, Shortcuts, and Settings.

The application menu (File, Edit, Library, Help) mirrors import, CSV, analyze, write tags, remove, settings, and shortcuts. On macOS, Settings sits in the app menu. On Windows and Linux it is also under File.

There is no breadcrumb, no footer of legal links, and no in-app page for donate or account. Unknown routes are not applicable. A bad `beat://` id returns an empty response from the protocol handler and the image or audio simply fails to load.

Back behavior: Escape closes the top dialog. There is no history stack.

3. Sitemap diagram



Every dialog returns to the same library window. The brand control is the only leave-the-app path.

4. User journeys

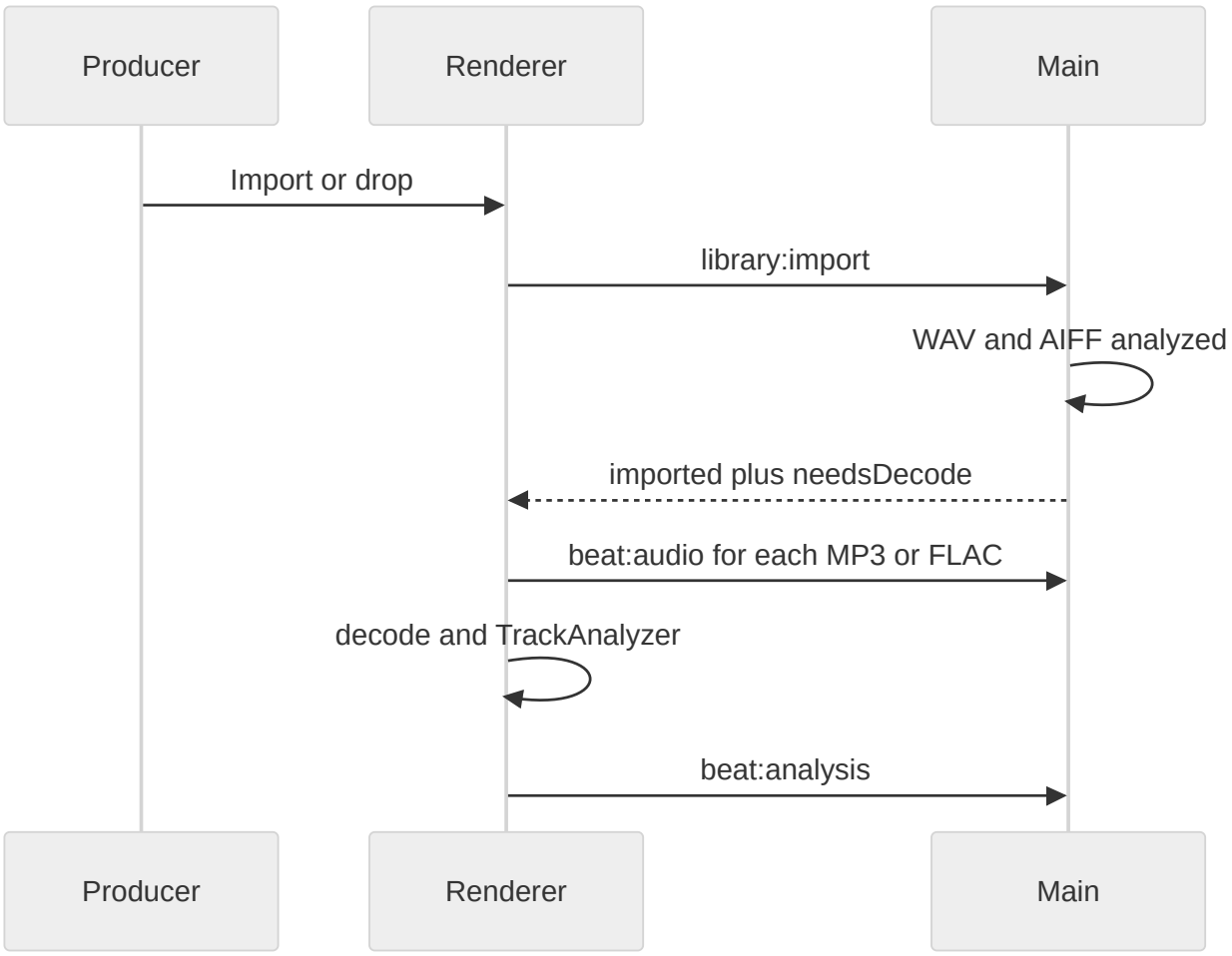
FLOW-001 First import

Trigger: empty library, or a new folder of bounces. Goal: rows with tempo and key.

1. The producer clicks Import, or File → Import Files…, or presses Ctrl/⌘ I. A file dialog filters to wav, mp3, aiff, aif, flac. Import Folder… (Ctrl/⌘ Shift I) picks a directory. Dropping files uses the same import.

2. `prep.importPaths` calls `library:import`. `Importer` walks the tree, skips duplicates and dotfiles, reads tags, and creates a `Beat`.
3. WAV and AIFF are analyzed in the main process before the call returns. MP3 and FLAC ids come back as `needsDecode`. The renderer reads bytes (`beat:audio`), decodes with `AudioContext.decodeAudioData`, runs `TrackAnalyzer`, and submits `beat:analysis`.
4. A sidecar JPEG named like the audio file is stored under `artwork/<id>.jpg`.
5. A toast uses `UserMessage.imported`. Example: "Imported 4 beats."

Success: rows show BPM and key. Failure: a file that throws is counted in "could not be read" and the others still import. An empty drop does nothing.



Import sequence. Local PCM is analyzed before the IPC returns. Compressed formats finish in the renderer.

FLOW-002 Read and correct analysis

Trigger: a row is selected. Goal: a working BPM and key the producer trusts.

1. Click the row, or use ↑ and ↓. The inspector shows the working BPM at two decimals, the key name, the Camelot badge, the detected line, confidence, and the waveform.
2. Half-time sets `bpmScale` to 0.5 and clears `bpmOverride`. Double-time sets 2. Use detected sets scale 1 and clears the override. Each scale is applied to `bpmDetected`, so half then half does not quarter the tempo.
3. Typing Working BPM sets `bpmOverride` if the number is between 40 and 300. Lock BPM sets `bpmLocked`. Key root and mode set `keyOverride` and `camelotOverride`. Lock key sets `keyLocked`.

4. Analyze (Ctrl/⌘ Enter) runs again. Locked fields stay. Unlocked fields update, and a successful analysis clears an unlocked override so the new detection shows.

Success: the table column matches `BpmMath.working` and `KeyChoice.working`. Failure: audio shorter than 0.75 s or under 8 kHz leaves BPM empty. Key confidence of 0 leaves the key empty. The idle line is "Analyze to measure tempo, key, and the waveform."

FLOW-003 Write tags and artwork

1. Edit fields in the inspector. Changes call `beat:update` immediately.
2. Add cover, or Adjust crop. Crop opens with zoom 1–4 in the slider (the math allows 1–6). Export size is 1500×1500 or 3000×3000. Use cover draws a JPEG at quality 0.92 and calls `beat:artwork`.
3. Write tags to file (Ctrl/⌘ S) calls `beat:write-tags` for the selection, or the active row if nothing is multi-selected. `TagCommit.fromBeat` builds the payload from working BPM, working key, settings producer name, and the artwork file.
4. Show in folder reveals the audio file. Export cover saves the JPEG via a save dialog.

Success toast: "Tags were written to the file." Failure toast: "Tags could not be written. The file may be open in another app or read-only."

FLOW-004 Catalog

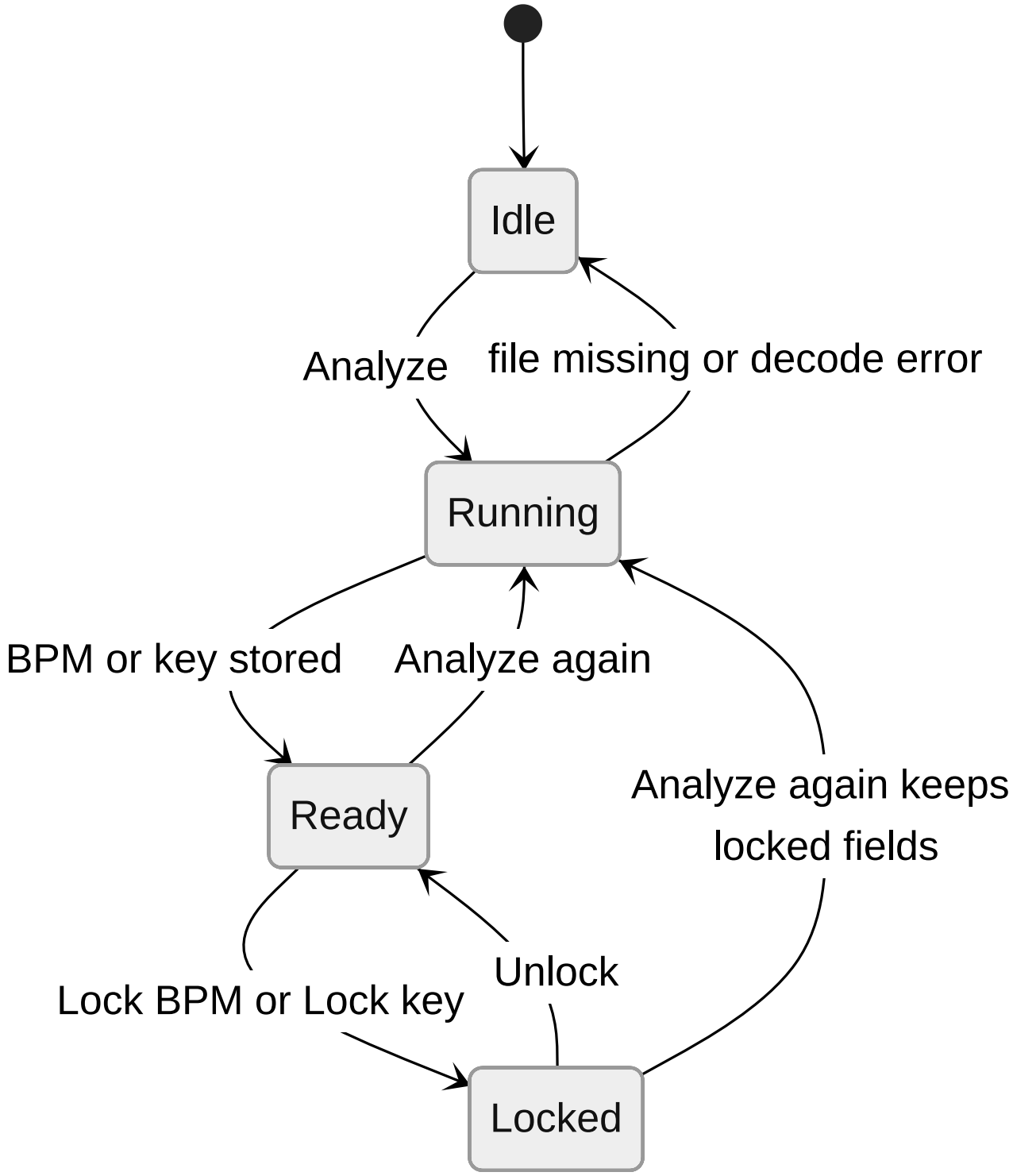
1. Filter in the rail. Sort by clicking Title, BPM, Key, Genre, Status, or Date. Default sort is imported, descending.
2. Shift-click selects a range. Ctrl or ⌘ click toggles a row. The header checkbox selects the visible rows.
3. Batch edit sends only the fields that are filled.
4. Export CSV (Ctrl/⌘ E) writes `CsvExport.render` for the selection, or all beats if the selection is empty (`library:csv` uses the full list when ids are empty).
5. Keys 1–5 set the active beat's status when focus is not in a field.
6. Remove asks for confirmation. Audio files stay on disk. The artwork JPEG in userData is deleted.

FLOW-005 Returning producer

Launch reads `library.json`. `missing` is recomputed from `fs.existsSync`. No network. The last settings (producer name, copyright template, export size) reload with the file.

5. Feature flows

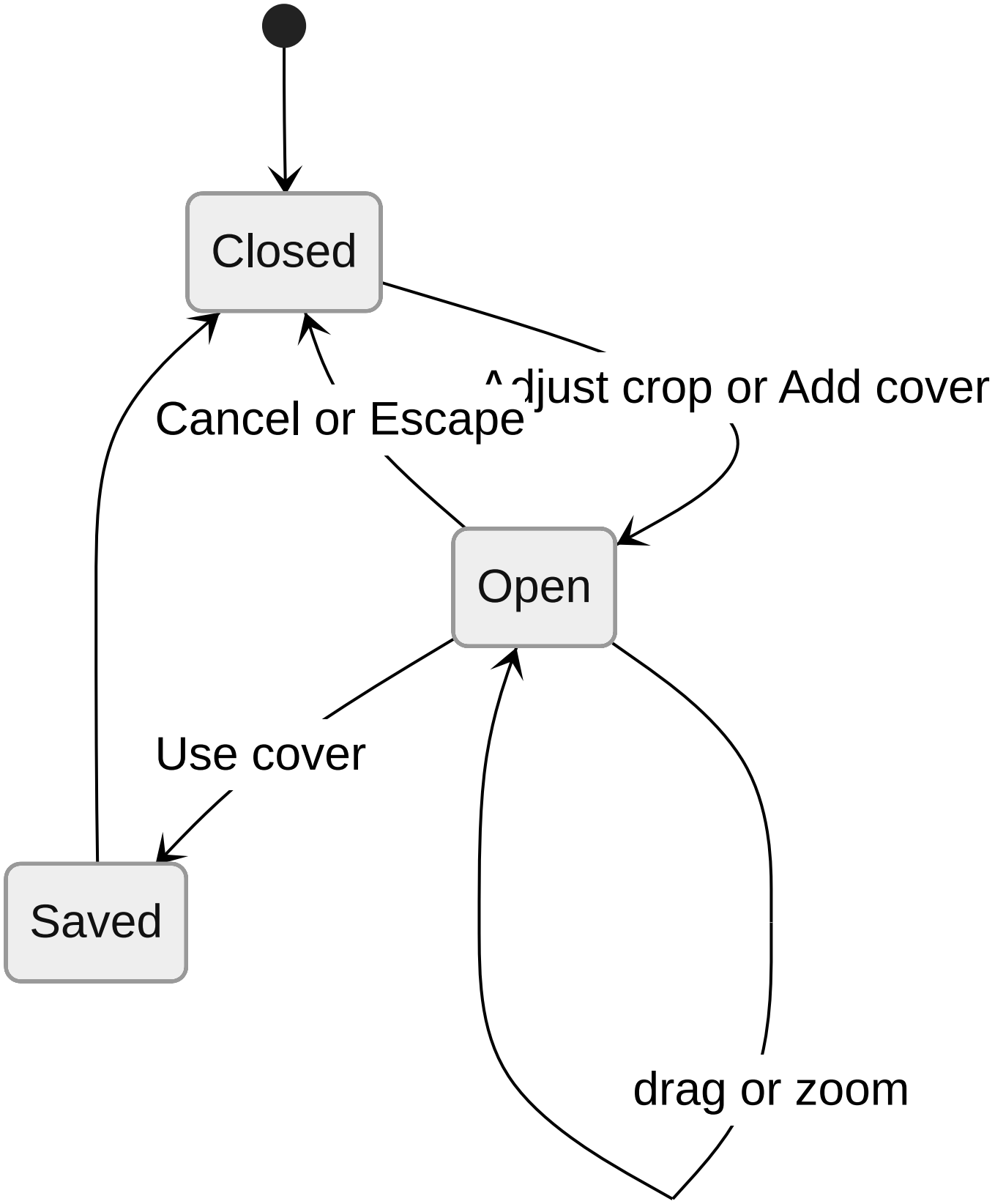
Analysis state



Analysis states for one beat. Lock is per field, so BPM can be locked while key still updates.

Half-time, double-time, and Use detected are enabled when a detected BPM exists and the BPM is not locked. The working BPM input is disabled while locked.

Crop state



Crop dialog. Use cover writes `artwork/<id>.jpg` and updates width and height on the beat.

Playback

Space toggles `audio#beat-player`, whose `src` is `beat://audio/<id>`. The play button does the same. Playback stops being the focus of the UI when another beat is selected because the element `src` changes with the beat.

Filters

`BeatQuery.match` requires every active filter. Search is a case-insensitive substring over title, artist, filename, genre, mood, key, Camelot, tags, notes, and comment. BPM filter uses the working BPM. Date filter compares the `date` string. Empty filters match everything.

6. Payment or checkout

Not applicable. The app does not start a payment, open a checkout, or talk to a payment provider. There is no error branch for a charge.

7. Error, empty, and edge states

State	What the producer sees
Empty library	"No beats in the library yet" and "Drop WAV, MP3, AIFF, or FLAC files here, or import a folder."
Filters exclude everything	"No beats match these filters."
File missing	Row is dimmed. Inspector: "This file is missing from disk." Write tags is disabled.
Bad ISRC	"ISRC must be 12 characters, like US-AWK-26-00001, or left blank." The previous value stays.
BPM outside 40–300	"BPM must be between 40 and 300."
Tag write failure	"Tags could not be written. The file may be open in another app or read-only."
Import failures	Toast ends with "N could not be read."
Duplicates	"N already in the library."
Cover smaller than 3000	"This image is smaller than 3000×3000. Exporting at 3000 will upscale it."
Offline	No change. Analysis does not use the network.
Slow disk	The row stays in the busy set until analyze returns. There is no separate timeout toast.
JavaScript disabled	Not applicable. The shell is Electron.
Unknown menu command	Ignored.

8. Notifications and messages

Toasts last 4.2 seconds (`App.tsx`). Exact strings live in `src/core/library/UserMessage.ts` and are quoted in section 7. Additional toasts:

Event	Text
CSV saved	`Catalog saved to` plus the path returned by the save dialog
Import	`UserMessage.imported`
Tags	`tagsWritten` or `tagsFailed`
Validation	The string returned by `beat:update` as `{ error }`

No email is sent. No push notification.

Menu labels: Import Files…, Import Folder…, Export Catalog CSV…, Analyze, Write Tags to File, Remove from Library, Settings, Keyboard Shortcuts, About.

9. Screenshots

Captured on the development Electron window (Linux, production build, 1440×900 class window) with a seeded library. The PNG files are pull-request artifacts, not files in the git tree:

Artifact	What it shows
`library_analysis.png`	Library with Night Drive selected. Working BPM 140.25, key F# minor, Camelot badge, detected tempo, key confidence, half-time and double-time controls.
`metadata_artwork.png`	Cover, embedded size, title Night Drive, producer Awkward J, genre, mood, tags.
`crop_3000.png`	Crop cover dialog with export size 3000×3000.

The seed beats used for those shots were generated locally (not the owner's catalog): Night Drive at 140.25 BPM in F# minor, plus three other loops. They are not shipped in the installer.

A phone-width screenshot is not a product surface. The 980px collapse is described in section 1.

10. Glossary

Term	Meaning
Rail	Left filter column.
Inspector	Right column for the selected beat.
needsDecode	MP3 or FLAC ids the renderer must analyze after import.
Working BPM	See ABP-PRD glossary.

11. Open questions

ID	Question	Owner	Blocked
Q-006	Support email, if any, and where it should appear	Awkward J	No contact line in the window today
Q-009	Should remove also offer to delete the audio file?	Awkward J	v1 never deletes audio

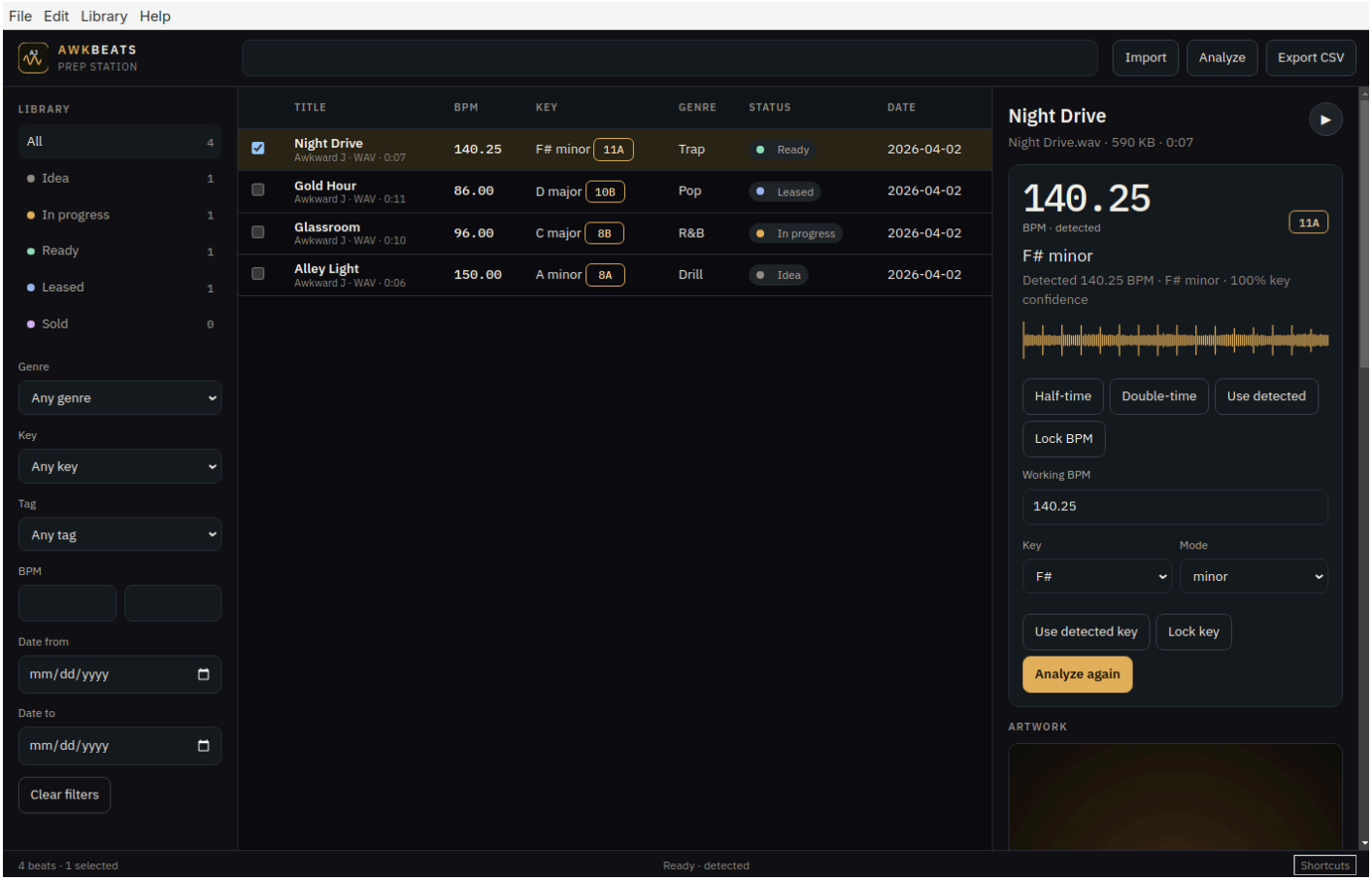
12. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Flow for the single-window 1.0.0 app.	Owner build brief.
2026-10-21	1.1	The title bar, empty library, no-match state, and About box use `build/logo.png`.	Owner supplied the transparent logo.

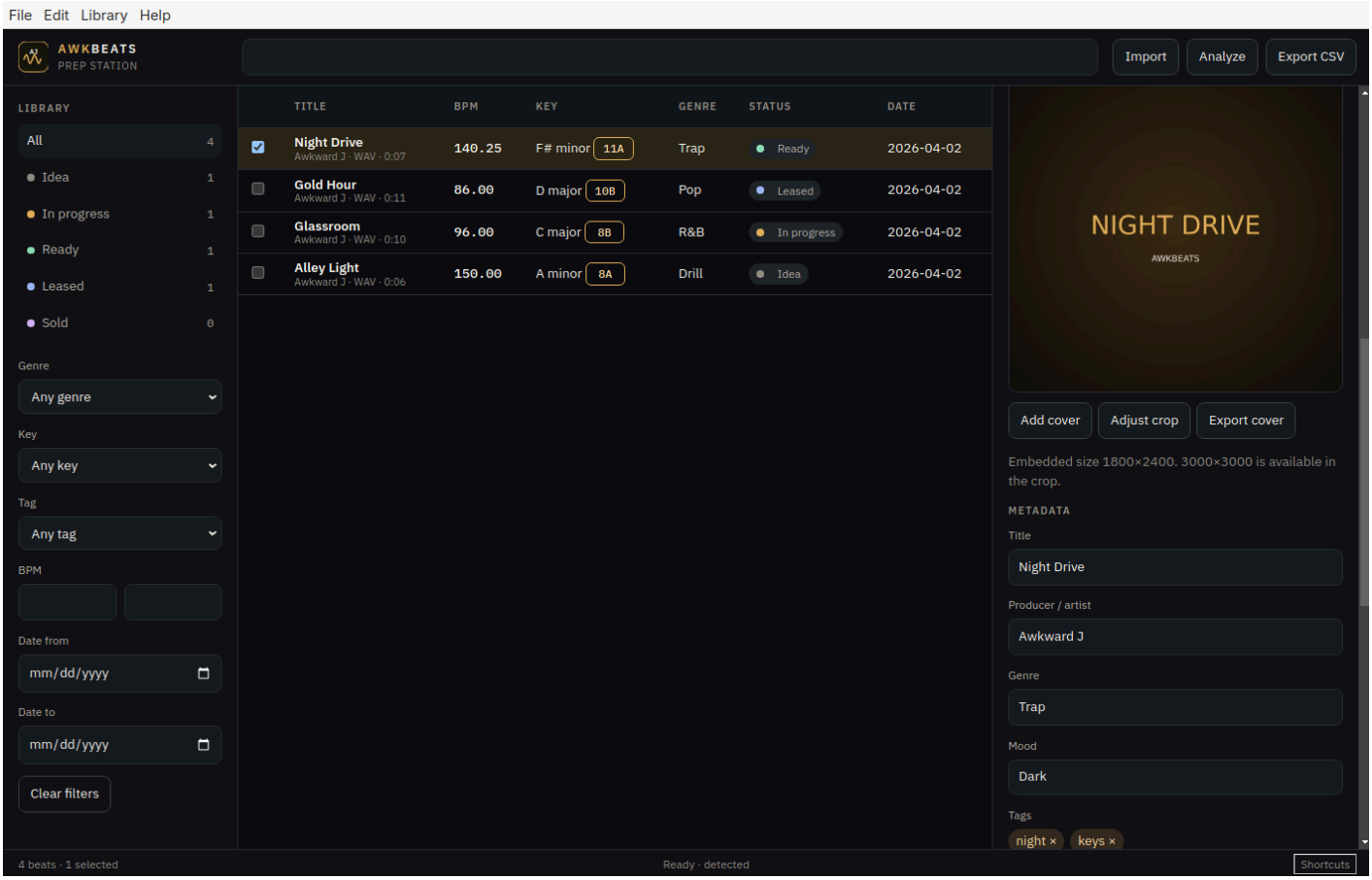
13. Sources

- `src/renderer/App.tsx`, `Inspector.tsx`, `Sidebar.tsx`, `TrackTable.tsx`, `dialogs.tsx`.
- `electron/main.ts` menu and IPC.
- `src/core/library/UserMessage.ts`, `Importer.ts`, `BeatRules.ts`.

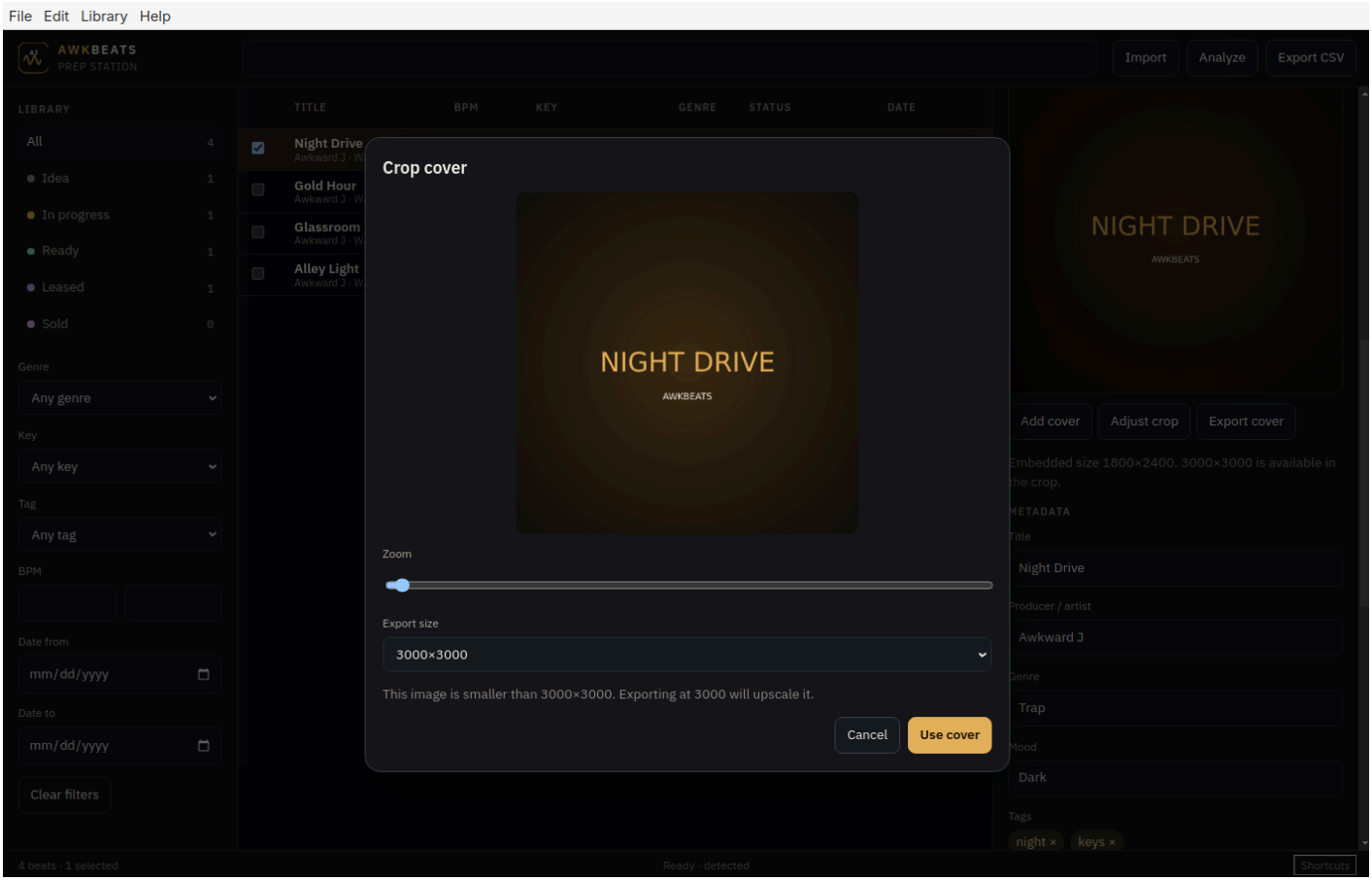
- Screenshots described in section 9, taken 2026-10-21.



Library and analysis. Night Drive selected, BPM and key visible.



Metadata and artwork editor.



Crop cover dialog with a 3000 export size.

Design brief — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	Design Brief
Document ID	ABP-DESIGN
Version	1.1
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Brand site: https://awkbeats.com
Operator	Awkward J (legal entity UNCONFIRMED)

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1. Brand

Name: AwkBeats. Product: Prep Station. Producer: Awkward J. Site: https://awkbeats.com.

Tagline: none in the product. The top bar subtitle is "Prep Station".

The interface should feel like a studio tool: dark, quiet, gold used for the brand and for the key badge, numbers in a monospace face so BPM columns line up.

Personality, each as a contrast the UI should hold:

- Precise, not playful. BPM is `140.25`, not "about 140".
- Calm, not nightclub. No animated gradients, no strobe on the waveform.
- Direct, not chatty. Buttons say "Write tags to file", not "Let's save your vibe".
- Local, not cloudy. No account avatar, no sync icon.

Audience reminder: a producer at a desk, often for hours. Density is welcome. Decoration is not.

2. Logo and mascot

There is no mascot. The mark is the owner's AwkBeats iOS icon: a gold A with a purple inner facet. Q-004 is resolved.

Asset	Path	Use
App icon	`build/icon.png` 1024×1024, opaque, black ground	Source for every installer target
macOS icon	`build/icon.icns`	`electron-builder.yml` `mac.icon` and `dmg.icon`. PNG chunks icp4, icp5, ic11, icp6, ic12, ic07, ic08, ic13, ic09, ic14, ic10
Windows icon	`build/icon.ico`	`win.icon`, NSIS installer and uninstaller. Sizes 16, 24, 32, 48, 64, 128, 256
Linux icons	`build/icons/<size>x<size>.png`	`linux.icon`. Sizes 16, 24, 32, 48, 64, 128, 256, 512, 1024
In-app mark	`build/logo.png` 1024×1024, RGBA	Title bar, empty states, about panel. `BrandMark` in `src/renderer/App.tsx` is 36×36. Empty state is 88×88

Clear space: keep at least a quarter of the mark's height empty around it in marketing. Do not recolor the gold or the purple, do not stretch it, and do not put the opaque black square in the title bar. The title bar and empty states use the transparent file so the A sits on `#0c0d10`. The retired AJ waveform is not in the repo.

There is no Open Graph image and no favicon separate from the app icon. Not applicable for a desktop window.

3. Color

Tokens are in `src/renderer/styles.css` on `:root`. Ratios were computed with the WCAG relative-luminance formula on 2026-10-21.

Token	Hex	RGB	Role
`--bg`	`#0c0d10`	12, 13, 16	Window background
`--bg-rail`	`#101114`	16, 17, 20	Top bar, rail, inspector, footer
`--bg-elev`	`#17191e`	23, 25, 30	Inputs, analysis card, buttons
`--bg-elev-2`	`#21242b`	33, 36, 43	Pills, play button
`--bg-active`	`#2a2418`	42, 36, 24	Selected table row
`--line`	`#2c3038`	44, 48, 56	Hairline borders
`--line-strong`	`#3c414b`	60, 65, 75	Button borders

Token	Hex	RGB	Role
`--text`	`#f4f0e8`	244, 240, 232	Primary text
`--text-dim`	`#b7b1a6`	183, 177, 166	Secondary text
`--text-faint`	`#938d85`	147, 141, 133	Labels, counts, section titles
`--gold`	`#e4b15a`	228, 177, 90	Accent, primary button
`--gold-2`	`#f3d7a4`	243, 215, 164	Gold text, focus
`--gold-dim`	`rgba(228,177,90,0.16)`	—	Tag chip fill
`--mint`	`#8ee0b8`	142, 224, 184	Status Ready
`--blue`	`#9eb6f0`	158, 182, 240	Status Leased
`--coral`	`#f09a86`	240, 154, 134	Destructive text
`--violet`	`#d2b0f0`	210, 176, 240	Status Sold
`--danger`	`#ff6d6d`	255, 109, 109	Reserved danger
`--focus`	`#f3d7a4`	243, 215, 164	Focus ring
Table header background	`#12141a`	18, 20, 26	Sticky header, not a token
Primary ink	`#1a1408`	26, 20, 8	Text on gold buttons
Modal surface	`#14161b`	20, 22, 27	Dialogs
Toast surface	`#242017`	36, 32, 23	Toasts

Pair	Ratio	WCAG
`--text` on `--bg`	17.10	AAA
`--text-dim` on `--bg`	9.12	AAA
`--text-faint` on `--bg`	5.68	AA
`--text-faint` on `--bg-rail`	5.52	AA
`--text-faint` on `#12141a`	5.39	AA
`--text-faint` on `--bg-elev`	5.14	AA
`--text-faint` on `--bg-active`	4.50	AA
`--gold` on `--bg`	9.93	AAA
`--gold-2` on `--bg`	13.94	AAA
`#1a1408` on `--gold`	9.36	AAA
`--mint` on `--bg`	12.50	AAA
`--blue` on `--bg`	9.63	AAA
`--coral` on `--bg`	8.95	AAA
`--danger` on `--bg`	7.09	AAA
`--text` on `--bg-active`	13.55	AAA
`--text` on `--bg-elev`	15.47	AAA

Status dots are 8px circles plus the status word. Gold means in progress, mint means ready, blue means leased, violet means sold, faint means idea. The word is always present, so a color-blind producer is not relying on the dot.

`--text-faint` was `#7e786e` during the first screenshots and measured 4.44:1 on `--bg` and 4.02:1 on `--bg-elev`, which misses AA for small text. It was changed to `#938d85` before the shipping commit. Section titles are 11px uppercase, so they need AA, not large-text AA.

4. Typography

Families, self-hosted from `@fontsource/ibm-plex-sans` and `@fontsource/ibm-plex-mono`, SIL Open Font License 1.1. Weights loaded in `src/renderer/main.tsx`: sans 400, 500, 600; mono 500. Fallback: `'Segoe UI', sans-serif` and `ui-monospace, monospace`.

Level	Size	Line height	Tracking	Weight	Where
Body	14px	1.45	0	400	`body`
Brand name	13px	—	0.14em	600	`.brand-name`
Subtitle	11px	—	0.08em	400, uppercase	`.brand-sub`
Section label	11px	—	0.12em	600, uppercase	`.section-label`, rail `h2`
Inspector title	20px	—	0	600	`.inspector-head h3`
Tempo	40px	1	-0.04em	500 mono	`.tempo`
Key	18px	—	0	400	`.key-line`
Table header	11px	—	0.08em	600, uppercase	`.tracks th`
BPM cells	14px	—	0	500 mono	`.mono`
Footer	12px	—	0	400	`.footer`
Empty title	22px	—	0	600	`.empty h2`
Dialog title	18px	—	0	600	`.modal h2`
Button	14px	—	0	400, primary 600	`.btn`

No responsive type scale. The window does not change font size with viewport.

5. Spacing, layout, and grid

The app grid is `60px` top bar, `minmax(0, 1fr)` workspace, `30px` footer. The workspace is `248px`, flexible table, `400px` inspector. Gap in the top bar is 16px. Rail padding is 16px. Table cells are 10px horizontal, row height 44px.

Breakpoint: `max-width: 980px` switches to two columns (see ABP-FLOW). There is no other breakpoint.

Spacing used repeatedly: 6, 8, 10, 12, 14, 16, 18px. There is no 8pt token file beyond the CSS itself. The window minimum is 1100×700.

6. Shape and depth

Token	Value	Use
`--radius`	12px	Analysis card
Inputs, buttons, status rows	8px	`.btn`, `.status-item`, fields
Camelot badge	6px	`.camelot`
Pills and chips	99px	`.pill`, `.chip`
Play	99px	`.play`
Modal	16px	`.modal`
Drop overlay	16px	`.drop`
Cover	10px	`.cover`
Toast	10px	`.toast`

Token	Value	Use
`--shadow`	`0 24px 60px rgba(0,0,0,0.45)`	Modal and toast
Focus	2px solid `--focus`, offset 2px	`:focus-visible`
Borders	1px `--line` or `--line-strong`	Panels and inputs

No stickers. No extra outline brand stroke.

7. Components

Code lives in `src/renderer/styles.css` and the TSX files named below.

Component	Anatomy	States	File
Brand button	32px mark, wordmark, subtitle	Default, focus-visible. Hover does not recolor.	`App.tsx`
Search field	Input, sr label	Default, focus	`App.tsx`
Button `.btn`	8×12 padding, 8px radius	Default, hover (gold border), disabled 0.45 opacity	CSS
Primary `.btn-primary`	Gold fill, ink text, weight 600	Hover `--gold-2`, disabled	CSS
Ghost `.btn-ghost`	No fill	Hover border	Footer
Danger `.btn-danger`	Coral text	Default	Remove
Status row	Dot, label, count	Default dim, hover and `.active` elevated	`Sidebar.tsx`
Table	Checkbox, title, BPM, key, genre, status, date	Hover `#191c22`, `.active` `--bg-active`, `.missing` faint text	`TrackTable.tsx`
Camelot badge	Mono code, gold border	Static	CSS `.camelot`
Analysis card	Tempo, key, buttons, waveform SVG	BPM lock disables half, double, input	`Inspector.tsx`
Cover well	Square, cover-size background	Empty faint label, or image	`Inspector.tsx`
Chip	Gold dim pill, remove	Default	Tags
Field	Label 12px dim, control	Error text `.error` in coral 12px	Inspector and dialogs
Modal	Scrim `rgba(0,0,0,0.62)`, panel `#14161b`, width min(520px, 100%), wide min(680px, 100%)	Open	`dialogs.tsx`
Toast	Gold border, max-width 360px, bottom right	Timed 4.2s	`App.tsx`
kbd	Mono, gold-2, 1px border, 2px bottom	Static	Shortcuts

8. Iconography and illustration

The only illustration is the AwkBeats A (gold with a purple facet) and the analysis waveform, which is a gold SVG path of the beat's 160 peaks (stroke `#e4b15a` via `--gold`, width 1.4, no fill). Status uses CSS dots, not an icon font. No third-party icon set.

9. Motion

No entrance animation is defined. Hover and color changes are instant except where the browser defaults a transition; the reduced-motion rule sets `transition: none` and `animation: none` on everything. The drop overlay appears and disappears with state, not a timed animation. There is no celebration effect.

10. Imagery and media

Covers are JPEG. PNG can be read for size (`ImageSize.ts`) and embedded if the bytes are PNG (`ArtworkPayload` allows `image/jpeg` or `image/png`). The crop dialog always writes JPEG. Export sizes are 1500 and 3000. Display size in the inspector is the column width, square, `object-fit` via CSS `background-size: cover`.

Alt text: the cover is decorative next to the beat title, which is a heading. The empty cover says "No cover". There is no separate alt attribute because the well is a div with a background image.

Audio is played through the `beat://` URL. Formats: wav, mp3, aiff, flac.

11. Voice and copy

Tone: adult, brief, specific. Reading level: plain English, short sentences. Use: beat, library, tag, cover, half-time, double-time, Camelot, producer. Avoid: placeholder, lorem, TODO, "simply", "just", "unlock your", emoji, exclamation marks.

Example of the rule already applied: the empty state names the four formats and the two actions (drop or import). It does not say "Your journey starts here."

The ISRC example `US-AWK-26-00001` is a shape sample in the error string. It is not asserted to be a registered code.

12. Accessibility design rules

- Focus ring: 2px `#f3d7a4`, offset 2px, on `:focus-visible` only.
- Contrast minimum: WCAG AA (4.5:1) for text that is not large. The pairs in section 3 meet that after the faint-color change.
- Do not encode status by color alone. The label is printed.
- Keyboard path must cover import, search, analyze, write, and status. It does (ABP-FLOW).
- Reduced motion must remove animation. It does.
- Hit targets: table rows are 44px. The play control is 36px. Rail rows are shorter. Do not shrink the table row.

13. References and inspiration

No external product was copied. The layout (filters, table, inspector) is the usual pattern of a desktop music librarian. What was taken as a principle: dark chrome and a single accent so the Camelot badge and the primary button are the loudest elements. What was not copied: skeuomorphic decks, neon VU meters, purple-on-black "AI" gradients, Inter/system-stack as the only face.

IBM Plex was chosen because the OFL license is compatible with a commercial app and the mono companion matches the sans.

14. Do and don't

Do:

1. Show BPM with two decimal places.
2. Show the key name and the Camelot code together.
3. Keep gold for the brand, the primary button, the Camelot badge, and the waveform.

4. Write button labels as verbs the producer already uses ("Write tags to file", "Half-time").
5. Leave the audio file where the producer saved it.

Don't:

1. Don't put developer notes, TODOs, or lorem in the window.
2. Don't use a light theme for v1. `color-scheme: dark` is set.
3. Don't recolor status dots without updating this table.
4. Don't add a second accent that competes with gold.
5. Don't watermark the cover image unless the owner asks. The producer name is a tag, not a stamp.
6. Don't replace IBM Plex with a font whose license forbids embedding in a sold app.

15. Screens

The running window was captured during development. Images are pull-request artifacts, not committed PNGs:

- Library and analysis: Night Drive selected, BPM 140.25, F# minor, half-time and double-time controls visible.
- Metadata and artwork: cover well, title, producer, genre, mood, tags.
- Crop dialog: title "Crop cover", export size 3000×3000, Use cover.

Annotated notes: the tempo figure is `.tempo` at 40px mono. The Camelot code sits in `.camelot`. The primary action in the inspector is "Write tags to file". The crop primary action is "Use cover".

16. Glossary

Term	Meaning
Gold	`#e4b15a`, the only brand accent.
Ink	`#1a1408`, text on gold buttons.
Faint	`#938d85`, metadata labels.

17. Open questions

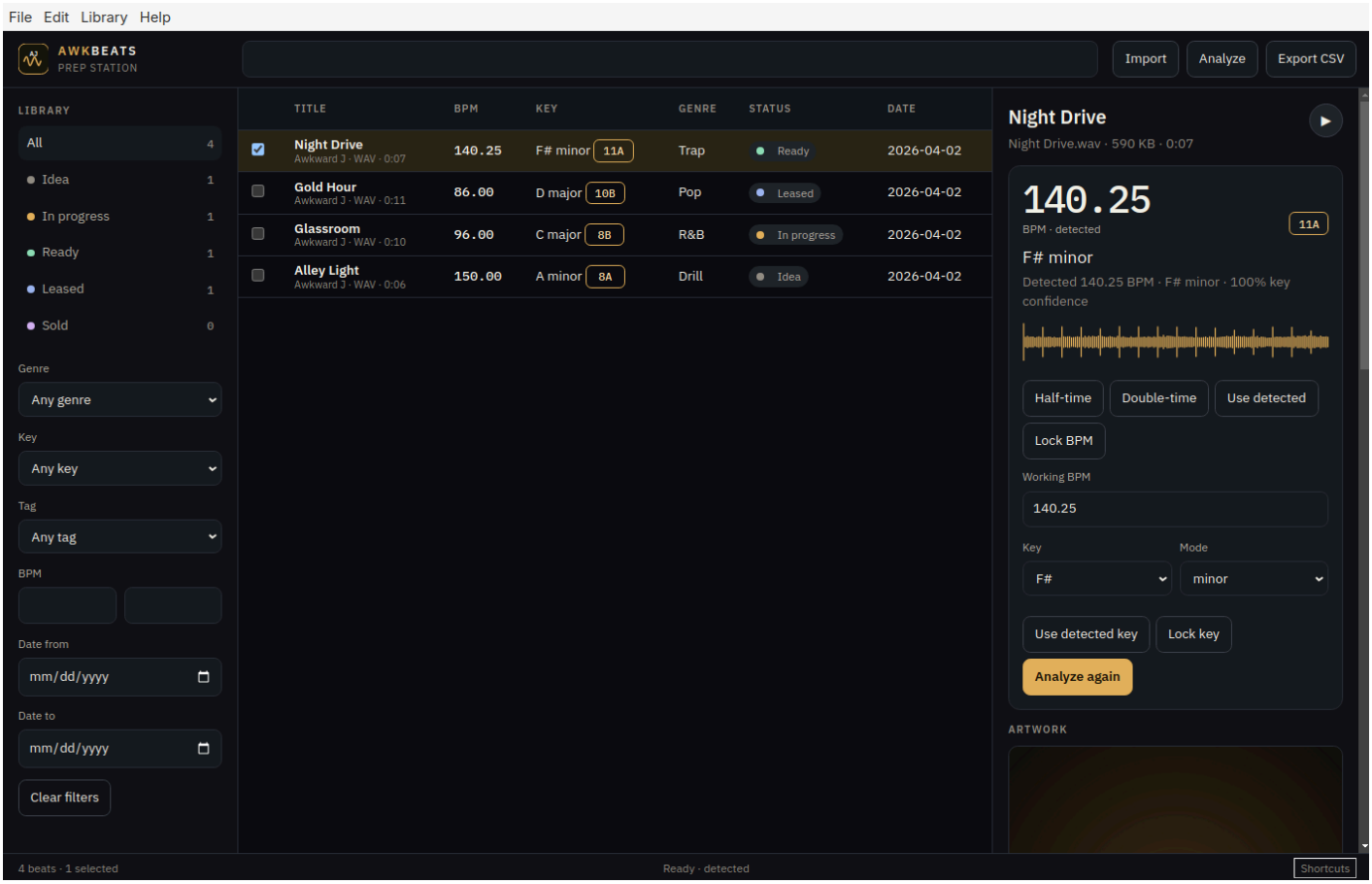
ID	Question	Owner	Blocked
Q-004	Resolved. The iOS icon and transparent logo are the marks in `build/`.	Awkward J	None
Q-010	Is a wordmark lockup needed beyond the top bar?	Awkward J	Marketing site, not this app

18. Changelog

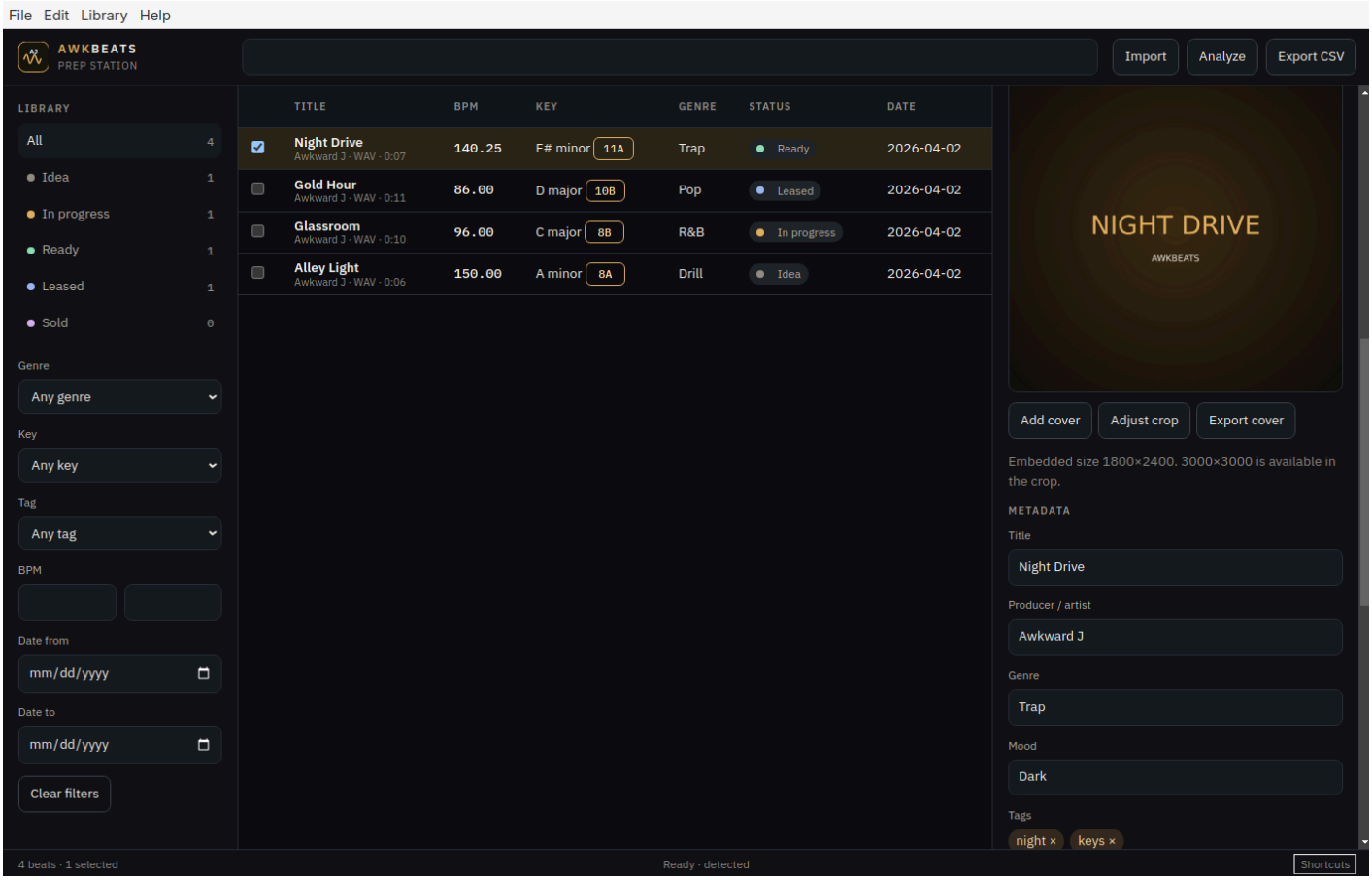
Date	Version	Change	Why
2026-10-21	1.0	Brief matched to `styles.css`, including the faint-color contrast fix.	Owner build brief. `#7e786e` failed AA for 11px labels.
2026-10-21	1.1	The mark is the iOS gold A. Transparent file in the window, opaque file on installers.	Owner supplied the logo.

19. Sources

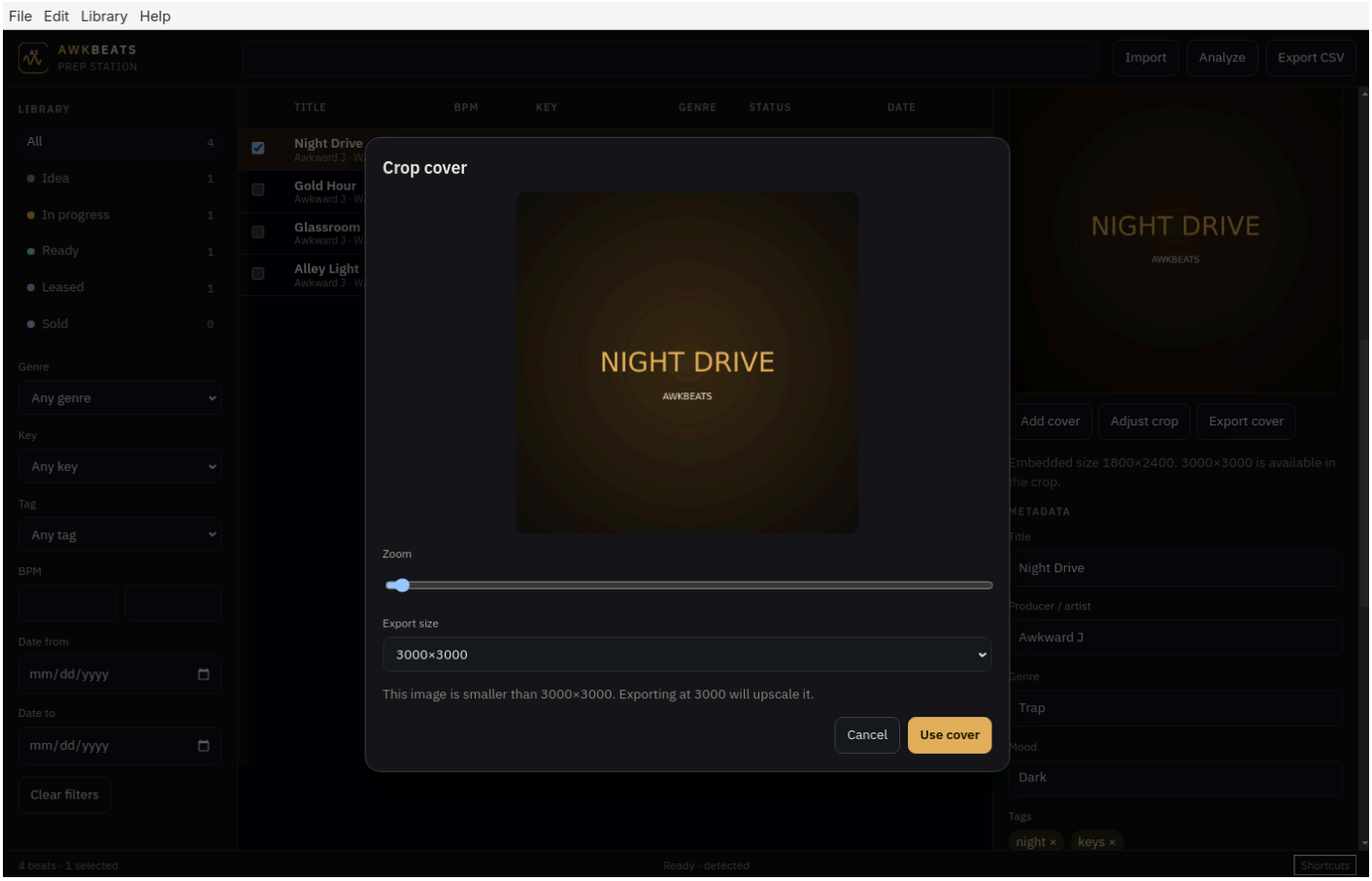
- `src/renderer/styles.css`, `main.tsx`, `App.tsx`, `Inspector.tsx`, `dialogs.tsx`.
- `build/icon.png`, `build/logo.png`, `build/icon.ico`, `build/icon.icns`, `build/icons/`.
- Contrast script run locally on 2026-10-21 (WCAG relative luminance).
- `@fontsource/ibm-plex-sans` and `@fontsource/ibm-plex-mono` license: SIL OFL 1.1.



Library and analysis. Night Drive selected, BPM and key visible.



Metadata and artwork editor.



Crop cover dialog with a 3000 export size.

Data schema — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	Backend and Data Schema
Document ID	ABP-DATA
Version	1.0
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Data stays on the producer machine.
Operator	Awkward J (legal entity UNCONFIRMED)

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1. Data architecture

There is no database server. Three places hold data:

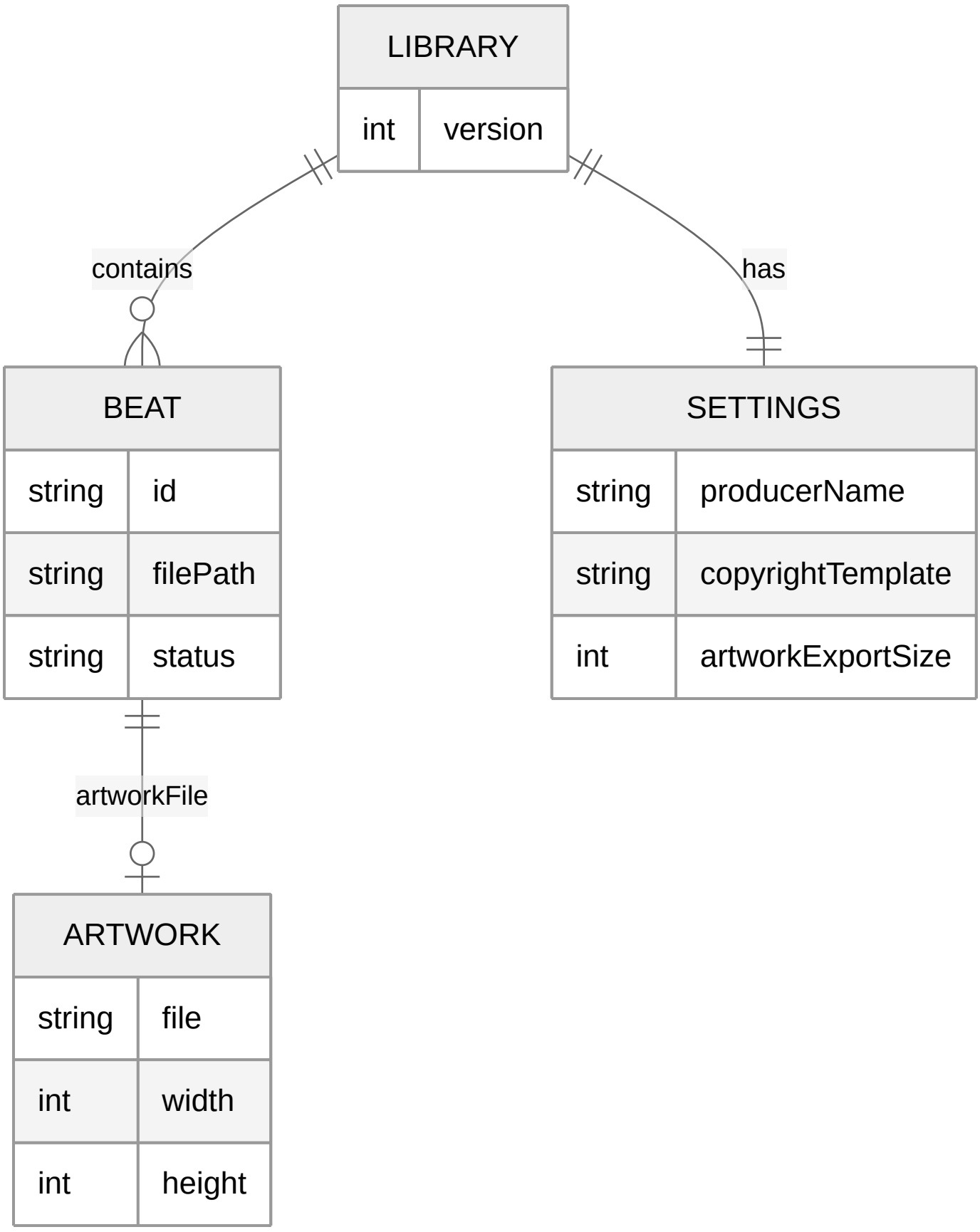
Data	Where
Catalog and settings	`library.json` in the Electron userData directory
Cover images	`userData/artwork/<beat id>.jpg`
Audio	The path the producer imported. Not copied.
UI state (selection, filters, sort)	React memory. Lost on quit.



The JSON catalog points at audio paths and at artwork files. The window's filters are not persisted.

Override userData with `AWKBEATS_USER_DATA`. The default path is Electron's for this app id (`com.awkbeats.prepstation`).

2. Entity diagram



One library file holds many beats and one settings object. Artwork is a sibling file, not a row inside the JSON.

3. Entities

`missing` is computed on read (`fs.existsSync`) and is not a reason to treat the file as a second source of truth. It is still present on the TypeScript type. `LibraryStore.save` writes the in-memory beats, which include whatever `missing` value they had when loaded. Callers should treat `missing` from `list()` as live and the stored boolean as stale. That is a known quirk (debt, low): the next save can persist the previous `missing` flag. `list()` overwrites it before return.

LibraryFile

Field	Type	Nullable	Default	Constraints	Description	Example	PII
version	number	no	1	must be 1 or load throws	Schema version	1	no
beats	Beat[]	no	[]	array	Catalog	[]	mixed
settings	Settings	no	DEFAULT_SETTINGS	object	App settings	see below	no

Settings

Field	Type	Nullable	Default	Constraints	Description	Example	PII
producerName	string	no	Awkward J	none	Written as artist fallback and as PRODUCER tag	Awkward J	name
copyrightTemplate	string	no	© {year} Awkward J	`{year}` replaced at import	Default copyright	© 2026 Awkward J	no
artworkExportSize	1500 or 3000	no	3000	only those two accepted in `updateSettings`	Default crop export	3000	no

Beat

Implemented in `src/core/library/types.ts`. `id` is `crypto.randomUUID()` from `BeatRecord.create`.

Field	Type	Nullable	Default	Constraints	Description	Example	PII
id	string	no	uuid	unique in practice	Primary id	a uuid	no
filePath	string	no	—	unique per library (`findByPath`)	Absolute audio path	/Users/me/beats/Night Drive.wav	path may contain a username
fileName	string	no	basename	—	Display name	Night Drive.wav	no
format	wav, mp3, aiff, flac	no	—	enum	Container	wav	no
fileSize	number	no	stat size	bytes, import max 2 GiB	Size	1440004	no
durationSec	number	yes	null	seconds	Duration	8.0	no
sampleRate	number	yes	null	Hz	Rate	44100	no
title	string	no	tag or basename	—	Title	Night Drive	no
artist	string	no	tag or producerName	—	Producer / artist	Awkward J	name
genre	string	no	""	free text, suggestions in `BeatCatalog.genres`	Genre	Trap	no

Field	Type	Nullable	Default	Constraints	Description	Example	PII
mood	string	no	""	suggestions in `BeatCatalog.moods`	Mood	Dark	no
tags	string[]	no	[]	max 24, each max 32, case-insensitive unique	Tags	night, keys	no
comment	string	no	""	max 4000 on update	Comment	—	could be personal
copyright	string	no	template	max 400 on update	Copyright line	© 2026 Awkward J	no
isrc	string	no	""	empty or 12-char canonical	ISRC	USAWK2600001	no
date	string	no	tag or mtime	stored as YYYY-MM-DD slice	Date	2026-10-21	no
bpmDetected	number	yes	null	0.01 resolution when set	Detector BPM	140.25	no
bpmScale	0.5, 1, or 2	no	1	enum	Half or double	1	no
bpmOverride	number	yes	null	40–300 if set	Hand BPM	null	no
bpmLocked	boolean	no	false	—	Lock	false	no
bpmConfidence	number	yes	null	0–1	Detector confidence	0.98	no
keyDetected	string	yes	null	Camelot name	Detected key	F# minor	no
camelotDetected	string	yes	null	like 11A	Detected code	11A	no
keyOverride	string	yes	null	—	Hand key	null	no
camelotOverride	string	yes	null	—	Hand code	null	no
keyLocked	boolean	no	false	—	Lock	false	no
keyConfidence	number	yes	null	0–1	Gap score	1	no
status	BeatStatus	no	idea	idea, in_progress, ready, leased, sold	Catalog status	ready	no
notes	string	no	""	—	Internal notes, not written to the audio tag	Hold for Friday	could be personal
artworkFile	string	yes	null	filename only	Cover filename	`<id>.jpg`	no
artworkWidth	number	yes	null	pixels	Width	1800	no
artworkHeight	number	yes	null	pixels	Height	2400	no
peaks	number[]	no	[]	160 values, 3 decimals, 0–1	Waveform	0.12	no
missing	boolean	no	false	recomputed on read	File absent	false	no
analyzedAt	string	yes	null	ISO timestamp	Last analysis	2026-10-21T00:00:00.000Z	no
createdAt	string	no	now	ISO	Created	same	no
updatedAt	string	no	now	ISO	Updated	same	no
importedAt	string	no	now	ISO	Imported	same	no

Enums

Beat status labels (`BeatStatusLabel`): Idea, In progress, Ready, Leased, Sold.

Genre suggestions: Trap, Drill, Hip-Hop, R&B, Pop, Afrobeats, Amapiano, Jersey Club, House, Techno, Boom Bap, Lo-fi, Rage, Dancehall, Reggaeton. The field accepts other strings.

Mood suggestions: Dark, Energetic, Chill, Melodic, Aggressive, Sad, Bouncy, Soulful, Spacey, Uplifting.

Key names come from `Camelot.roots`: C, C#, D, D#, E, F, F#, G, G#, A, A#, B, plus `major` or `minor`. Flats parse onto the sharp spelling (`Eb major` is `D# major`).

Tag payload written into audio

`TagCommit.fromBeat` maps the working BPM and working key, plus `settings.producerName` as `producer`. ID3 frames (`TagWriter.id3Tags`):

App field	ID3	Vorbis	Notes
title	TIT2	TITLE	
artist	TPE1	ARTIST	
genre	TCON	GENRE	
comment	COMM eng	COMMENT	
bpm	TBPM with two decimals	BPM	
key	TKEY up to 3 chars (`F#m`) plus TXXX KEY	KEY and INITIALKEY (Camelot)	TKEY cannot hold "F# minor"
camelot	TXXX CAMELOT	CAMELOT	
tags	TXXX TAGS, comma-separated	TAGS	
mood	TXXX MOOD	MOOD	
copyright	TCOP if it can be made to start with a year, plus TXXX COPYRIGHT	COPYRIGHT	`TagText.copyrightFrame`
isrc	TSRC	ISRC	
date	TDRL and TYER when the first four chars are a year	DATE	
producer	TXXX PRODUCER	PRODUCER	from settings, not the beat artist field
artwork	APIC type 3	PICTURE type 3	

WAV also writes RIFF LIST/INFO (INAM, IART, IGNR, ICMT, ICRD, ICOP). Existing INFO and `id3 ` chunks are replaced. Other RIFF chunks, including `data`, stay. AIFF writes NAME, AUTH, ANNO, and ID3, replacing older copies of those chunks.

FLAC keeps STREAMINFO and audio frames, drops old VORBIS_COMMENT (type 4), PICTURE (type 6), and PADDING (type 1), then writes a new comment, an optional picture, and 256 bytes of padding. Vendor string: `AwkBeats Prep Station`.

CSV columns

`CsvExport.headers`: id, title, artist, file_name, file_path, format, duration_sec, bpm, bpm_detected, bpm_scale, bpm_override, bpm_locked, key, camelot, key_detected, camelot_detected, key_locked, key_confidence, genre, mood, tags (joined with `|`), status, notes, comment, copyright, isrc, date, analyzed_at, imported_at, updated_at. RFC 4180 quoting.

4. Client-side storage

localStorage, sessionStorage, IndexedDB, and cookies are not used.

The only persisted client state is `library.json` and the artwork directory, written by the main process. Shape: `LibraryFile` above. Version strategy: `load()` throws if `version !== 1`. There is no migrator yet. Size: one JSON rewrite of the full catalog per change. Cleared by deleting the userData folder, or by Remove for individual beats (artwork file deleted, audio kept).

5. IPC reference

These are not HTTP endpoints. They are `ipcMain.handle` channels. Auth is the local user. No rate limit. The preload method is the name in `PrepApi`.

ID	Channel	Preload	Purpose	Payload	Success	Error	Side effects	File
API-001	library:list	list	All beats	none	Beat[]	throw on bad JSON at load	recomputes missing	main.ts
API-002	settings:get	settings	Read settings	none	Settings	—	none	main.ts
API-003	settings:update	updateSettings	Patch settings	Partial Settings	Settings	ignores illegal export size	writes JSON	LibraryStore
API-004	library:import	importPaths	Import	string[] paths	ImportResult	per-file messages inside result	writes JSON, may write artwork	Importer
API-005	dialog:audio	pickAudio	Open files	none	string[]	[] if cancelled	none	main.ts
API-006	dialog:folder	pickFolder	Open folder	none	string or null	null if cancelled	none	main.ts
API-007	dialog:artwork	pickArtwork	Open image	none	{name, bytes} or null	null if cancelled	none	main.ts
API-008	beat:update	updateBeat	Patch one beat	id, UserPatch	{beat} or {error}	error string, no throw for validation	writes JSON	LibraryStore
API-009	beat:batch	batchUpdate	Patch many	ids, BatchPatch	Beat[]	—	writes JSON	LibraryStore
API-010	beat:remove	remove	Remove rows	ids	Beat[]	—	deletes artwork files, writes JSON	LibraryStore
API-011	beat:analyze	analyze	Analyze WAV/AIFF	ids	{beats, needsDecode, errors}	errors[] for failures	writes JSON	main.ts
API-012	beat:analysis	submitAnalysis	Store renderer analysis	id, AnalysisResult	{beat} or {error}	—	writes JSON, respects locks	LibraryStore.applyAnalysis
API-013	beat:audio	readAudio	Whole file bytes	id	Uint8Array	throws if missing	none	main.ts
API-014	beat:write-tags	writeTags	Write audio tags	ids	{id, ok, message}[]	message on failure	mutates the audio file	TagWriter
API-015	beat:artwork	saveArtwork	Store JPEG	id, bytes, width, height	Beat	—	writes jpg and JSON	LibraryStore
API-016	beat:export-artwork	exportArtwork	Save dialog	id	path or null	null if none	writes a copy	main.ts
API-017	library:csv	exportCsv	Save CSV	ids (empty means all)	path or null	null if cancelled	writes the CSV the user chose	CsvExport
API-018	beat:reveal	showInFolder	Show in folder	id	void	—	OS shell	shell.showItemInFolder

ID	Channel	Preload	Purpose	Payload	Success	Error	Side effects	File
API-019	app:site	openSite	Open brand site	none	void	—	external browser	https://awkbeats.com only
API-020	app:startup	takeStartup	MP3/FLAC imported via `--import`	none	string[] ids, then cleared	—	none after read	main.ts
API-021	menu-command	onMenu	Menu to renderer	command string	event	—	none	main.ts
API-022	beat://audio/id and beat://cover/id	audio and img src	Stream file	id in URL	file bytes	empty if unknown	none	protocol.handle

Example `beat:update` validation error body: `{ "error": "BPM must be between 40 and 300." }`.

`--import <path>` on the executable runs import before the window loads and queues compressed ids for `app:startup`.

Idempotency: import of an existing path is a duplicate skip. Analyze overwrites unlocked analysis fields. Tag write replaces prior ID3, INFO, or Vorbis comment blocks.

6. Third-party API calls made by the server

Not applicable. The main process does not call a remote API. `shell.openExternal('https://awkbeats.com')` is a browser handoff, not an API call. music-metadata and node-id3 are in-process.

7. Webhooks

Not applicable. None incoming or outgoing.

8. Background jobs and scheduled tasks

Not applicable. No cron, no queue. Analysis runs on the IPC call stack (main for WAV/AIFF, renderer for MP3/FLAC). A long file is capped at 90 seconds of samples inside `TrackAnalyzer`.

9. File and object storage

Path	Permissions	Contents	Cleanup
userData/library.json	user umask	Catalog	Replaced atomically via `library.json.tmp` then rename
userData/artwork/*.jpg	user umask	Cover	Deleted in `LibraryStore.remove`
Audio path on the beat	unchanged	Original audio	Never deleted by the app
CSV path from the save dialog	user choice	Export	Not tracked
Exported cover from the save dialog	user choice	JPEG copy	Not tracked

No sweep of old files. Orphan artwork can remain if a beat id is removed from the JSON by hand. The app's remove path deletes the jpg it knows about.

10. Data lifecycle

Retention: until the producer deletes the files. No backup is taken by the app. Restore: copy `library.json` and `artwork/` back into userData. Audio must still be at `filePath`. If a path changed, the row shows missing; v1 has no relink dialog.

Deletion: Remove drops the row and the artwork. It does not delete audio. Uninstalling the app does not delete userData on every OS; Electron's default is to leave it. The producer deletes that folder to wipe the catalog.

11. Migrations and seeding

No migration runner. A future version must bump `version` and teach `load()` to upgrade. Rollback of a bad JSON is the `.tmp` file only if a crash happens before rename; after rename, the previous file is gone. Keep a copy if you experiment.

Seeding: `scripts/seed-demo.ts` is a development tool. It is not invoked by the app or by CI. It writes generated loops, tags, and a cover into a directory you pass. Do not point it at a real catalog. `tests/fixtures/silence.flac` is a 0.1 second silent FLAC used only by tests.

12. Security of data

No encryption at rest beyond what the operating system does for the user account. No passwords, no hashes. Access control is the OS user. The renderer cannot read `library.json` except through IPC. Tag writes require a library id, and the path written is the path stored on that beat, not a path chosen by the renderer in the write call.

`beat:audio` returns the full file. That is intended for local decode. Do not expose the preload bridge to untrusted web content. Navigation guards in `electron/main.ts` exist for that reason.

13. Glossary

Term	Meaning
userData	Electron application data directory.
Working BPM	`bpmOverride` else `bpmDetected * bpmScale`.
TXXX	ID3 user-text frame. Used for MOOD, TAGS, KEY, CAMELOT, PRODUCER, COPYRIGHT.
APIC	ID3 attached picture.

14. Open questions

ID	Question	Owner	Blocked
Q-003	Validate ISRC against a real registrant prefix?	Awkward J	Stricter than the 12-character pattern
Q-011	Should notes be written into the audio comment as well?	Awkward J	v1 keeps notes only in the catalog
Q-012	Relink when a file moves?	Awkward J	v1 marks the row missing

15. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Schema for library version 1 and the IPC list.	Owner build brief.

16. Sources

- `src/core/library/types.ts`, `LibraryStore.ts`, `CsvExport.ts`, `Importer.ts`, `AudioFiles.ts`, `BeatRules.ts`.
- `src/core/tags/TagWriter.ts`, `TagCommit.ts`, `TagTypes.ts`, `TagReader.ts`.
- `electron/main.ts`, `electron/preload.ts`.

Implementation plan — AwkBeats Prep Station

Document control

Field	Value
Project	AwkBeats Prep Station
Document title	Implementation Plan
Document ID	ABP-PLAN
Version	1.1
Status	Final (as shipped)
Last updated	2026-10-21 (UTC)
Author	Cursor Agent, for owner Awkward J
Repo	https://github.com/Awkward8787/awkbeats-prep-station
Branch	`cursor/awkbeats-prep-station-e2e2`
Commit SHA	f99d3c6a38ad256e123fa13b50e0f7660369bd61
Live URL	None. Installers are GitHub Actions artifacts, not a Release.
Operator	Awkward J (legal entity UNCONFIRMED)

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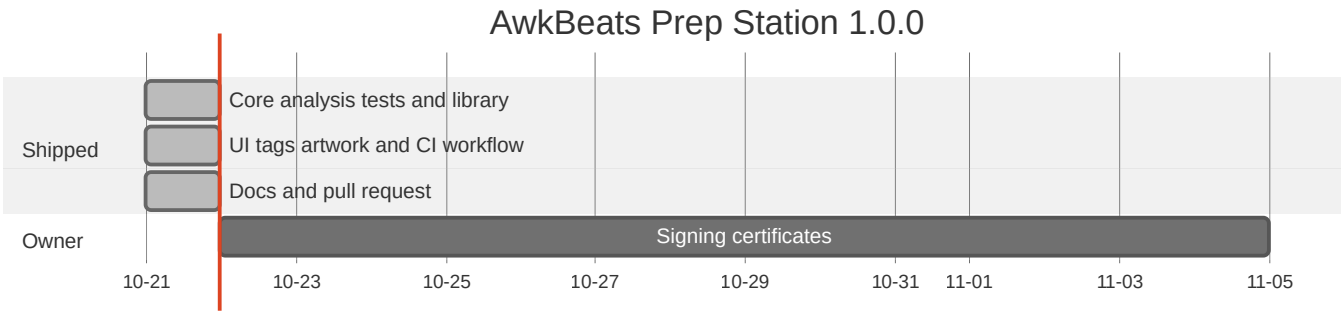
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1. Delivery summary

AwkBeats Prep Station 1.0.0 was built in this repository as a local Electron app: library, offline BPM and key analysis, tag writing, cover art, CSV, batch tools, tests, and a GitHub Actions workflow that builds an unsigned macOS dmg and a Windows exe. No GitHub Release is published. The builder was Cursor Agent. The owner is Awkward J. Status: shipped as source and as CI installer artifacts once the shipping commit's workflow is green.

2. Phases and milestones

Dates are the day the tree was produced (2026-10-21). They are not estimates of future work.



The build landed in one working tree. The signing bar is the owner's, and the two weeks on the chart are a placeholder span, not a promise.

Phase	Status	Result
Analysis, library, tags, tests	Shipped	`src/core`, `tests/`
UI and installers workflow	Shipped	`src/renderer`, `.github/workflows/installers.yml`
Documentation and PDFs	Shipped	`docs/`
Signed release	Planned	Blocked on certificates. Not started.

3. Work breakdown

ID	Task	Phase	Requirements	Status	Notes
TASK-001	Original BPM and key detectors, no GPL library	Build	FR-004, FR-005, NFR-001	Shipped	`src/core/analysis`
TASK-002	Reference-tone tests at 0.01 BPM and 24 keys	Build	NFR-002	Shipped	`tests/bpm-key.test.ts`
TASK-003	Library JSON, import, filters, CSV, locks	Build	FR-001–FR-003, FR-006, FR-011, FR-012	Shipped	`src/core/library`
TASK-004	ID3, RIFF INFO, Vorbis, artwork embed	Build	FR-008, FR-009	Shipped	`TagWriter.ts`
TASK-005	Dark UI, batch, crop, shortcuts	Build	FR-007, FR-013–FR-017, NFR-004	Shipped	`src/renderer`
TASK-006	License check in CI	Build	NFR-001	Shipped	`scripts/check-licenses.mjs`
TASK-007	Actions workflow for dmg and exe	Build	NFR-003	Shipped	artifacts, `--publish never`
TASK-008	Contrast fix for faint text	Build	NFR-008	Shipped	`--text-faint #938d85`
TASK-009	Six docs and PDFs	Docs	—	Shipped	this folder
TASK-010	Apple notarization	Owner	NFR-003 follow-up	Planned	Secrets in README
TASK-011	Windows Authenticode	Owner	NFR-003 follow-up	Planned	EV recommended
TASK-012	Confirm legal entity and license	Owner	Q-001, Q-002	Planned	LICENSE is proprietary pending confirmation
TASK-013	Use the iOS app icon and transparent logo	Owner	Q-004	Shipped	`build/icon.png`, `build/logo.png`. AJ svg removed
TASK-014	Do not publish a Release	Build	brief	Shipped	`publish: null`

The pull request that contains this file is the delivery PR into `main`. Commit SHAs are on that branch.

4. Review gates

Review	Date	Verdict	Findings	Resolution
Unit tests	2026-10-21	Pass required before commit	BPM and tag assertions kept strict	Shipped with the tests green
License scan	2026-10-21	Pass	Production tree had no GPL-family license	`npm run licenses`
Typecheck	2026-10-21	Pass	`tsc --noEmit`	Shipped
Contrast	2026-10-21	Fail, then pass	`#7e786e` was 4.02:1 on elevated surfaces	Changed to `#938d85`, minimum 4.50:1
External code review, accessibility audit, security audit	—	Not run	No third-party report was attached to the brief	Deferred. Debt is listed in ABP-TRD.
UI check on Electron	2026-10-21	Exercised	Library, analysis card, metadata, crop dialog captured	Screenshots are PR artifacts

No finding was waived by weakening a test.

5. Testing and QA

Automated: `npm test`, `npm run typecheck`, `npm run licenses`. What each test file covers is listed in ABP-TRD section 13.

Manual script used on the Linux Electron build:

1. Launch with `AWKBEATS_USER_DATA` pointed at a scratch directory.
2. Import four generated loops (WAV). Confirm the table lists them.
3. Select the 140.25 BPM F# minor loop. Confirm the inspector shows 140.25, F# minor, and a Camelot badge.
4. Scroll to cover, title, producer, genre, mood, and tags.
5. Open Adjust crop. Confirm the export size control offers 3000×3000.
6. Half-time and double-time are unit-tested (`BpmMath.half(140.25)` is 70.13, and the scale is always applied to the detected tempo). The buttons call `save({ bpmScale: 0.5 | 2, bpmOverride: null })`.

Devices: development GUI on Linux with Electron 33.4.11. macOS and Windows are covered by the installer jobs compiling, not by a human clicking the packaged app in this session. Browsers: not applicable.

Results: unit tests and typecheck passed on the shipping tree. Installer artifact names are confirmed when the Actions run for this commit is green. If that run fails, the fix lands on the same branch before the PR is treated as done.

6. Launch checklist

Item	State	Evidence
Domain and DNS for the app	Not applicable	No app host. awkbeats.com is the brand site and is not changed by this repo.
SSL for the app	Not applicable	Local app
Secrets for unsigned build	Done	None required
Signing secrets	Not done	Owner
Payments	Not applicable	No provider
Legal pages inside the app	Not done	Not in scope for v1. Sale terms are Q-005.
Analytics	Done as none	No SDK

Item	State	Evidence
CSP	Done	`src/renderer/index.html`
404 page	Not applicable	No router
App icon	Done	`build/icon.png`, `build/icon.ico`
Performance	Done as measured	ABP-TRD section 9
Accessibility contrast	Done	ABP-DESIGN section 3
Backups	Not done by the app	Producer copies userData
Monitoring	Not applicable	No service
Installers	Workflow shipped	Artifact upload steps
Release published	Not done, on purpose	`--publish never`

7. Deployment runbook

There is no server deploy.

To produce installers from a desktop:

1. Install Node 22.
2. `npm ci`
3. On a Mac: `npm run dist:mac`. On Windows: `npm run dist:win`.
4. Take the files in `release/`. Do not run electron-builder publish.

From GitHub:

1. Push the branch. Workflow `Build installers` starts on every push.
2. Open the run. Download `AwkBeats-Prep-Station-macOS` and `AwkBeats-Prep-Station-Windows`.
3. Expected names: `AwkBeats-Prep-Station-1.0.0-mac-arm64.dmg`, `AwkBeats-Prep-Station-1.0.0-mac-x64.dmg`, `AwkBeats-Prep-Station-1.0.0-win-x64.exe`.

Verify: tests job green, both artifact jobs green, files non-empty. Rollback: download the previous run's artifacts. Access: whoever can push to the repo can start a build. Signing secrets, once added, should be environment secrets limited to maintainers.

Do not create a GitHub Release from this workflow.

8. Post-launch and owner actions

No temporary files need deleting in the repo. `dist/` and `release/` are gitignored.

Owner actions, in order:

1. **Apple signing and notarization.** Enroll the legal entity in the Apple Developer Program. Create a Developer ID Application certificate. Export the `.p12`. Create an App Store Connect API key. Add GitHub secrets `CSC_LINK` (base64 p12), `CSC_KEY_PASSWORD`, `APPLE_API_KEY` (base64 p8), `APPLE_API_KEY_ID`, `APPLE_API_ISSUER`, `APPLE_TEAM_ID`. Then set `mac.identity`, `hardenedRuntime: true`, and notarization in `electron-builder.yml`. Details are in the README.
2. **Windows code signing.** Buy an OV or EV certificate. EV is what SmartScreen trusts without a reputation warmup. Add `WIN_CSC_LINK` and `WIN_CSC_KEY_PASSWORD` (or `CSC_LINK` / `CSC_KEY_PASSWORD`). `signAndEditExecutable` is already true.

3. **Confirm the legal entity** for the copyright string. The app currently says Awkward J. Q-001.
4. **Confirm the license.** The repo ships `LICENSE` as all rights reserved because the app may be sold. Say if that should change. Q-002.
5. **ISRC.** The check is 12 characters. Provide a registrant prefix only if you want a tighter check. Q-003.
6. **Brand.** Done. The iOS icon is the installer art. The transparent logo is the in-app mark. Q-004.
7. **Distribution.** Decide where installers are downloaded. Do not use a GitHub Release until you ask for one. This delivery does not publish one.
8. **Support email.** None is shown. Send one if it should appear. Q-006.

9. Roadmap and next phases

ID	Task	Requirements	Depends on	Status	Size
TASK-010	Signed macOS build	NFR-003	Q-001, Apple secrets	Planned	One workflow change after secrets exist
TASK-011	Signed Windows build	NFR-003	Certificate	Planned	Same
TASK-015	Relink missing files	FR-020	Q-012	Planned	Small dialog
TASK-016	Optional image watermark	FR-010	Owner asks	Deferred	Not in v1
TASK-017	SQLite catalog	—	Catalog growth	Deferred	New store behind `LibraryStore`
TASK-018	Linux package	—	Owner asks	Deferred	electron-builder linux target
TASK-019	Focus trap on dialogs	NFR-007	—	Planned	`dialogs.tsx`
TASK-020	Price and storefront	Monetization	Q-005	Deferred	No number was given. Do not invent one.

"Size" is structural, not a schedule.

10. Risks and mitigations

Risk	Mitigation
CI cannot sign, so the first artifacts warn on open	Documented. Unsigned is the accepted v1 state.
A future contributor adds essentia or another GPL detector	`npm run licenses` fails the test job
Detector wrong on a dense mix	Override, lock, half-time, double-time. Tests stay strict on generated material.
`GITHUB_TOKEN` tempts electron-builder to publish	`publish: null` and `--publish never`

11. Maintenance plan

- Dependencies: review npm updates when touching the app. Re-run tests and `npm run licenses` after any audio or tag dependency bump. Do not upgrade music-metadata past a major version without re-checking decimal BPM, because 7.14.0's common reader drops the fraction and `TagReader.readNative` compensates.
- Secrets: only exist once the owner adds them. Rotate by replacing the GitHub secret and revoking the old certificate or API key at the vendor. No rotation is due today.
- Certificates: Apple and Windows certs expire on the vendor's calendar. Renew before the signed job starts failing.
- Content: genre and mood lists are in `BeatCatalog`. UI strings are in `UserMessage.ts` and the components.
- Cadence: re-run the installer workflow on each push. No separate staging cron.
- Docs: bump the document version and regenerate `docs/pdf` when behavior changes.

12. Glossary

Term	Meaning
Artifact	A file uploaded by GitHub Actions, downloaded from the workflow run. Not a GitHub Release.
Developer ID	Apple certificate for apps distributed outside the Mac App Store.
EV certificate	Extended Validation code-signing certificate for Windows.

13. Open questions

The open questions from the other documents apply here. The ones that block owner work:

ID	Question	Owner	Blocked
Q-001	Legal entity	Awkward J	Apple certificate name
Q-002	Keep the proprietary license?	Awkward J	LICENSE
Q-003	ISRC prefix	Awkward J	Tighter validation
Q-004	Resolved. iOS icon and transparent logo are in `build/`	Awkward J	None
Q-005	Sell or not, price, store	Awkward J	Monetization. No price in the brief.
Q-006	Support email	Awkward J	Footer or About
Q-007	OV or EV Windows certificate	Awkward J	SmartScreen
Q-009	Should remove ever delete audio?	Awkward J	A second confirm. Default remains no.

14. Changelog

Date	Version	Change	Why
2026-10-21	1.0	Plan for the 1.0.0 delivery, including owner signing steps and an explicit decision not to publish a Release.	Owner build brief.
2026-10-21	1.1	TASK-013 shipped. The AJ svg is removed.	Owner supplied the iOS icon and transparent logo.

15. Sources

- `.github/workflows/installers.yml`, `electron-builder.yml`, `README.md`, `LICENSE`.
- `package.json` scripts `dist:mac`, `dist:win`, `licenses`, `test`.
- ABP-PRD, ABP-TRD, ABP-FLOW, ABP-DESIGN, ABP-DATA in this folder.
- Owner brief: desktop app, permissive analysis license, dmg and exe artifacts, no release publish.